

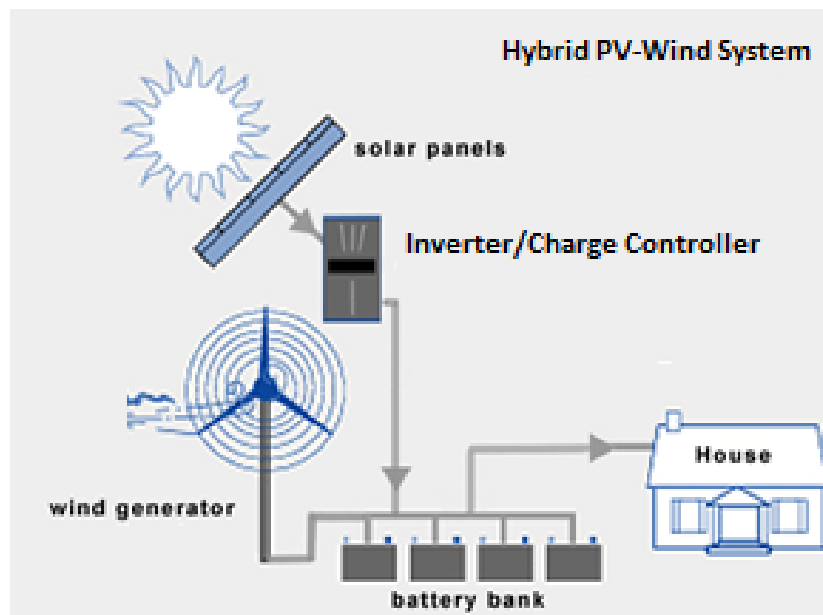


**KTH Industrial Engineering
and Management**

Feasibility Study of Solar-Wind Hybrid Power System for Rural Electrification at the Estatuene Locality in Mozambique

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Master of Science Thesis

KTH School of Industrial Engineering and Management

Energy Technology EGI-2015-033MSC EKV1089

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ABSTRACT

This project work focuses on the feasibility study of a hybrid PV-Wind System for rural electrification at the Estatuene Locality in southern Mozambique. This is in line with electricity network expansion, which, in Mozambique shows high implementation cost and low operation cost. Through field research, an analysis was made of the actual electrical demand in the Estatuene rural community. The wind data was collected from the installed weather stations in the region while the solar data were extracted internally from the HOMER software by introducing the site coordinates.

All the configurations, simulations and selection of hybrid systems were also made using HOMER. For the Estatuene rural community it was estimated a scaled annual average demand of 9.4 kWh/day with a peak load of 1.4 kW for DC charge; and a total scaled annual average of 133 kWh/day with a peak load of 15.3 kW for AC Charge. The annual mean solar potential is 5.205 kWh/m²/d, and the mean wind speed is 4.84 m/s for 12 meters above the ground. Thus the calculations and the selection of the best configuration of the hybrid system were crossed out with the technical specifications and costs of photovoltaic panels, wind turbines, power converter, batteries, and the electricity network, specifically for the comparison between an optimum hybrid system solution and two separate ones. The calculations presented an analysis of the technical and the financial viability of the selected hybrid system for local electric power production.

Keywords: hybrid system, solar power, PV, wind power, rural electrification, computational simulation, feasibility analysis, sustainable development.

ACKNOWLEDGMENT

It's our pleasure to thank to all SEE professors that gave us assistance during the course and transmitted their knowledge shaping us what we are today. Special thanks goes to our scientific supervisor's, Geraldo Nhumaio (Phd) and Miroslav Petrov (Phd) for their assistance during the conception, revision, corrections of this thesis and especially for understanding, patience and acceptance of our limitations as students. Our family for their patience and acceptance along the time that we weren't able to be with them. Our gratitude to Estatuene local authorities for their time and patience during community data surveys. Thanks to Dra. Miquelina Menenses (FUNAE, CEO) for her authorization and letting us to use FUNAE meteorological Data.

Thanks to God for giving us health and energy to work on this thesis.

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NOMENCLATURE

Symbol	Description	Unit
A	Cross section area of the cylinder	[m ²]
C _p	Betz limit Constant	---
E	Electric field	[N/c]
E _g	Semiconductor Energy Band gap	[N/c]
f	Frequency	[Hz]
F (V)	Weibull probability distribution function	---
G	Solar irradiance	[W/m]
I	Electric current feeding the load	[A]
I ₀	Reverse Diode current	[A]
I _d	Diod current	[A]
I _{MPP}	Current at maximum power point	[A]
I _{sc}	Short circuit current	[A]
J _D	Diffusion Current	[A]
K	Boltzman Constant	[J/k]
k	Shape factor	---
L	Length of the cylinder	[m]
$\dot{m}$	Air mass flow	[kg/s]
m	Diode ideality factor	---
η	Conversion efficiency	---
n, p types	Semiconductor materials	---
n _n	Electron concentration	---
n _p	Hole concentration	---
P	Power	[W]
P _{MPP}	Power at maximum power point	[Wp]
P _s	Power density	[W]
q	Electron charge	[c]
R	Blade radius	[m]
R _p	Parallel resistance	[Ω]
R _s	Series Resistance	[Ω]
T	Absolute temperature	[°K]
T _a	Ambient temperature	[°K]
T _c	Cell temperature	[°K]
v	Velocity	[m/s]
V(z)	Wind speed at height z	[m/s]
V _d	Diode voltage	[V]
V _{MPP}	Voltage at maximum power point	[V]
v _o	Oncoming wind speed	[m/s]
V _{oc}	Open circuit voltage	[V]
Z _o	Surface roughness length	[m]
α	Terrain roughness	---
β	Tilt angle of solar collector	[°Degree]
γ	Solar altitude angle	[°Degree]
θ	Rotor tip speed	[m/s]
λ_o	Tip speed ratio	---
ϱ	Air flow density	[kg/m ³]
ω	Angular velocity	[rad/s]

ACRONYMS AND ABBREVIATIONS

AC - Alternate current	PV-Photovoltaic
AGM – Absorbed glass mat	Sc – Scenarios
Ah – Ampere- hour	SC - solar cell
AWEA – American Wind Energy Association	SCR - Silicon controlled rectifiers
CdTe - Cadmium telluride	Si - Silicon
CIF – Cost, Insurance and Freight	TSR - Tip speed ratio
CNELEC - National Electricity Council	TW - Terawatt
DC - Direct current	TWp – Terawatt pic
DoD – Depth of discharge	USD – United States Dollar
EDM - Electricidade de Moçambique	VAWTS - Vertical Axis Wind Turbines
FET - field effect transistors	WECS - Wind Energy Conversion System
FF - fill factor	Wp – Watt pic
FUNAE- Fundo de Energia	WPD - wind power density
GaAs - Galium Arsenide	WT - Wind turbines
Ge - Germanium	
GSM - Global system for mobile communication	
Gwh – Gigawatt hour	
H ₂ O – Water	
H ₂ SO ₄ – Sulphuric acid	
HAWTS - Horizontal Axis Wind Turbine	
HCB - Hidroeléctrica de Cahora Bassa	
HOMER -Hybrid Optimization Model for Electric Renewable	
INAM - National Institute of Meteorology	
INE - Instituto Nacional de Estatística	
InP - Indium Phosphide	
km/h - kilometers per hour	
kWh – Kilowatt-hour	
LCOE - Levelized cost of electricity	
m/s– Meters per Second	
MoE - Ministry of Energy	
MPP - Maximal power point	
MSW – Municipal Solid Waste	
MW - Megawatt	
n/a – Not applicable	
NG - National grid	
NOCT - Normal Operating Cell Temperature	
NPC- Net present cost	
NREL - National Renewable Energy Laboratory	
O&M - Operation and maintenance	
OPE - Operating cost of Electricity	
p types - semiconductor materials	
P _b – Lead	
PbO ₂ – Lead oxide	
PbSO ₄ – Lead Sulfate	
PDF - Probability distribution function	

1 CHAPTER ONE: INTRODUCTION

1.1 Background to the study

At present, the increasing consumption of electricity drives the development of different forms of energy use around the world. This demand for electricity has been supplied mostly by fossil sources of energy. With high impact on the world economy caused by the increasing price fluctuations of fossil fuels, due to geopolitical issues and/or environmental disasters, the search for solutions that promote the sustainability of the current lifestyle of societies is growing in importance, which can bridge the rapidly growing energy demand in emerging economies such as Mozambique.

In Mozambique, the electricity supply is mainly realized through transmission lines, thus creating difficulties in meeting the distant or remote regions that due to orographic characteristics of places where people live, or geographical isolation, are not yet connected to the conventional national grid, as the population density to be supplied is low and do not justify large investments that represents the grid expansion, costs of network lines and power distribution maintenance.

The remote regions of the country are basically powered by isolated generator set systems. Despite of having relatively low cost, these systems are not the best solution for remote areas as they require a regular maintenance program that is rarely executed, leading to increased number of failures of the generators, which is reflected in a high rate of unavailability factors. This situation is also due to the increased fuel costs, especially regarding distant areas with a high shipping cost, which inflates the cost of electricity delivered by these systems.

Thus, this work aims to study the feasibility of a wind-PV hybrid system for local electricity production in order to power rural communities and to determine the circumstances in which a system of this nature becomes economically feasible for a specific site in rural Mozambique.

1.2 Statement of the Problem

Mozambique is a country located on the African Continent, which has 10 provinces and 129 districts. The total population is estimated to be approximately 22 million, with about 80% living in rural or suburban areas, grouped in small communities spread all over the country (INE, 2007) and their domestic primary energy needs are fully satisfied on the basis of firewood and charcoal (Cuamba, et al., 2006).

The country is rich in modern energy resources, however, more than 80% of the country's population is not connected to the national grid because of their location (rural areas) and poor economy, lack of basic infrastructure, high initial investment and costs of the transmission and distribution lines from the central

locations to remote areas which are considerably high. This situation leads to many remote areas remaining out of the reach of the centralized electrical grid long into the future (Hankins, 2009)

The electricity industry in Mozambique is presently represented by two major companies that include the hydropower producer “Hidroeléctrica de Cahora Bassa” (HCB) and the centralized electricity utility “Electricidade de Moçambique” (EDM). HCB generates over 90% of the country’s electricity at Cahora Bassa Dam and exports the bulk of this to South Africa. EDM has some limited generation capacity and buys most of its electricity from HCB. EDM is responsible for the management, maintenance and development of the national grid, including transmission and distribution lines. (Hankins, 2009)

The Mozambican Government set up in 1998 a new institution, named FUNAE (Fundo Nacional de Energia), to solve problems of off-grid energy demand and rural electrification. Since then FUNAE in the context of its supervision and financing of energy related projects, had the most impact on renewable energy development and initiative in promoting the use of stand-alone PV systems to supply electricity to small villages, community schools, health centres and local governmental services.

FUNAE is installing small power plants of primarily stand-alone PV systems with 4kW in small villages to electrify surrounding areas (usually within 300 meters) such as health centers, schools, police stations and households; and also have awarded a consultancy for a company “Gesto Energy” (Portuguese consultant) to map out the existing renewable energy sources - Solar, Wind, Hydro, Biomass, Municipal Solid Waste (MSW), Geothermal and Wave energy - over the country in order to prioritize the best renewable projects in Mozambique. Under the auspices of this consultancy some equipment to collect wind and solar data were installed over the country, whose readings will be used in the present study.

The combination of solar PV and Wind turbine hybrid systems is new in Mozambique (not yet implemented) and from the literature it’s seen that hybrids can be more cost effective than PV-alone or Wind-alone systems (Pazmino, 2007) by reducing the energy storage requirements. The present work attempts to investigate the possibility of providing electricity from Wind-Solar hybrid power system to a remotely located village that is outside of the main grid.

1.3 Objectives

1.3.1 General Objectives

- To study the feasibility of solar-wind hybrid power systems for rural electrification in Mozambique.

1.3.2 Specific Objectives

- To analyse and evaluate the renewable energy potential mainly for those two resources, solar and wind, at the Estatuene Locality in southern Mozambique;

- To make the prefeasibility analysis and estimate the load demand;
- To assess the technical feasibility of a hybrid solar-wind power system to meet the load requirements of the specific remote village electrification.
- To evaluate a strategy to optimize the size of the energy generation and storage subsystems;
- To assess techno-economic approach for determination of a system that guarantees the energy supply by the hybrid system with a lowest investment; and
- To analyse the effect of load size or load variation and at what extent the combination of the hybrid energy systems can reduce the costs if compared to a PV-only or Wind-only standalone off-grid system.

1.4 Methodology and Structure of the Thesis

This report is divided into five chapters. Chapter One is an introduction to the study, where the problem, justification, the aim and objectives of the study are outlined. Chapter Two is devoted to a theoretical discussion and literature review about the means of harnessing solar radiation and wind resources for energy applications, and their use as resource for energy production and leveraging them to generate electricity. Some techniques are also explored for the estimation of the available and extractable energy from these renewable sources and the factors that may affect their potential. The chapter is finalized with presentation of the energy profile of Mozambique.

Chapter Three presents the modelling approach used in this study by describing all the methodological procedures taken in place; then followed by Chapter Four where the results of the simulations and analysis are brought up and discussed, for identifying the best option for electrification of the chosen rural community of Estatuene.

Finally, in Chapter Five some important conclusions are drawn, together with general recommendations for future research and practical implementation.

The thesis work was carried out by a team of two students: Berino Silinto and Nelso Bila. The team performed literature review in books, papers, internet sources etc, where: Nelso were responsible for surveying information regarding solar resource, storage systems, inverters; and Berino for wind resource, hybrid systems, Mozambique energy profile. Both students worked together on the surveys of technical data and equipment characteristics (for wind turbines, photovoltaic panels, battery banks for electrical energy storage, converters/charge controllers, etc.), data analyses, discussion and results.

2 CHAPTER TWO: LITERATURE REVIEW

2.1 The Solar Energy Resource

The sun is the main energy source which is responsible for supporting all life activity around the world, such as the Earth's thermal comfort, photosynthesis in plants and the whole biogeochemical system. The sun emits its energy in form of electromagnetic radiation and after reaching the earth surface it is converted to other types of energy sources and used for many purposes.

The human beings are using the energy from the Sun in two main ways, i.e. for photo-electric generation and thermal conversion. These applications represent one big leap for the solution of the world energy shortage. For example, it is estimated that out of 1.76×10^{15} TW of raw solar energy striking the Earth, 60 TW can be economically converted into electricity and, considering that the estimation of the world energy demand until 2050 is about 25-30 TW, it is clear that only solar energy is enough to supply all demand and to free the world of fossil fuels (Kalogirou, 2009).

2.1.1 Thermal Conversion

The methods of thermal conversion of solar energy are based on absorption of radiant energy by black body surface. This can be a complex process, which varies according to the type of absorbent material. Involves diffusion photon absorption, electron acceleration, multiple collisions, but the final effect is the heating, or radiant energy of all qualities are transformed into heat represented by the increase in temperature. For instance, collectors can be used to gather solar radiation to produce temperature high enough to be used directly or further converted to electricity via thermomechanical processes such as for example a steam turbine cycle (Chen, 2011).

2.1.2 Photoelectric Conversion

This fundamental photoelectric conversion consists of the escape of electrons (electric current) from the clear metal surface when the light with certain frequency strikes on this surface.

2.1.3 The Photovoltaic System

A PV system is a composition of all the devices used to convert solar photons directly into electricity which are: Solar panel, storage unit, charge/discharge regulator and inverter if necessary to convert direct current (DC) to alternating current (AC). In some cases, depending on the purpose storage might not be necessary (e.g. connection to the grid). The key element of a PV system is the solar cell. This element is responsible for the conversion of solar radiation into electricity and its function is based on the photoelectric effect¹ that consists on electrical generation by certain materials when exposed to the light.

¹) Discovered by French physicist Edmond Becquerel in 1839

The first silicon solar cell (SC) was discovered by a French physicist, Edmond Becquerel in 1839. Becquerel experiments showed that certain materials produce a small amount of electricity when exposed to the light. This effect was firstly studied in metals such as silicon with performance of about 2%. The research proceeded and in 1954 was achieved a silicon solar cell with an efficiency of about 6%, reported by Chapin, Fuller and Pearson (Chen, 2011). Regarding its application, the SC's were firstly used to charge batteries of the United States Satellite (U.S. Vanguard) in 1958 (Maini, et al., 2011).

Due to high costs, the SC's were initially used only for space, military and scientific research purposes. However, with the energy crisis starting in the 1970's, interest emerged in developing of SC's for civilian purposes (Kalogirou, 2009).

2.1.4 Solar Cell Architecture

The SC's can be manufactured using different types of semiconductor materials, such as Silicon (Si), Germanium (Ge), Indium Phosphide (InP), Gallium Arsenide (GaAs), Cadmium telluride (CdTe), etc., but some of these materials (e.g. InP, GaAs) are not abundant in the Earth and therefore are much expensive if compared to those such as Si and Ge which leads to the prevailing usage of Si in commercial applications. The Ge is not much used because of its lower efficiency as can be seen in figure 2.1 below.

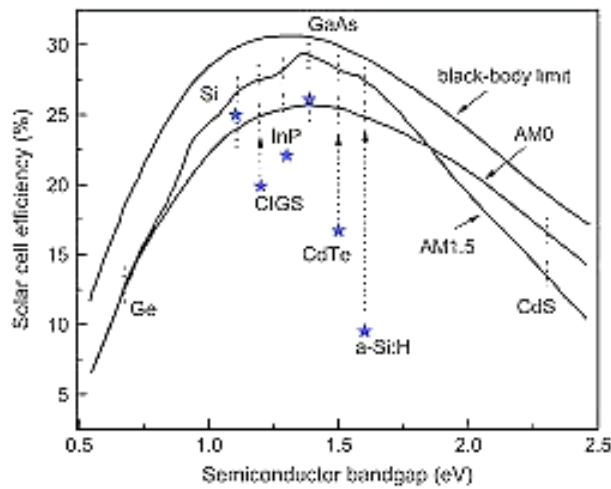


Figure 2. 1: Efficiency potential of some semiconductor materials. Source: (LEM, 2014)

2.1.5 Operation of Silicon Cells

SC is a junction of two types of materials, i.e. the **n** and **p** types of semiconductor materials. When those are joined together, free electrons in the n-type material move to the p-type material and also free holes from p-type material move to n-type material resembling the flow of electric charges, creating the so-called diffusion current (J_D). (Kalogirou, 2009).

When the charge carriers (electrons and holes) move from one side to another, they leave behind both donor and acceptor ions on their previous material (UNICAMP, 2014). Those left ions create spatial charges and, consequently there is an electric potential given by the following expression.

$$V = \frac{KT}{q} \ln \frac{n_n}{n_p} \quad (1)$$

Where n_n is the electron concentration and n_p is the holes concentration, K - Boltzmann constant ($1.38 * 10^{-23} \frac{J}{K}$), q - Electron charge ($1.6 * 10^{-19} C$) and T is the absolute temperature given in [°K]. As electric field is gradient of electric potential, it yields to the following expression (Luque, et al., 2011) and (UNICAMP, 2014).

$$E = \frac{KT}{q} \frac{1}{n} \nabla n \quad (2)$$

Where $\mathbf{E}$ is the Electric field given by [Newton/Coulomb].

The electric field due to the existence of spatial charges on a p-n junction, leads into the drift current (“diffusion force” on holes) with opposite direction to the diffusion current (“diffusion force” on electrons). Thus, the appearance of drift current stops the diffusion of charge carriers from one side to another and the junction works as a dielectric as is shown in figure 2.2.

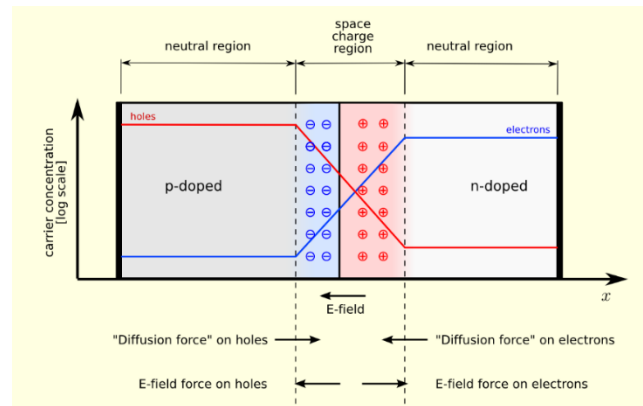


Figure 2. 2: P-N junction. Source: (TheNoise, 2007)

The electric field built up on junction that is yielded within the electric potential, disables the flux of charge carriers through it. However, it is of extreme importance to direct the charge carriers when they reach the depletion zone after they are excited by an external source (i.e. sunlight or thermal excitation).

The theory of light set up two important conclusions that instead of electromagnetic wave is also a joint of particles called photons. So, if those photons fall within the p-n junction, different situations such as electron-hole pair creation and recombination can take place according to their energy if compared to the semiconductor band gap (E_g).

If the energy of the photon that falls within the p-n junction is equal or more than E_g , after transferring this energy to the electron, it can jump into the conduction band and, due to electric dipole in depletion zone, electron can be directed to n-type material and the hole, to the p-type material (Yu, et al., 2010)

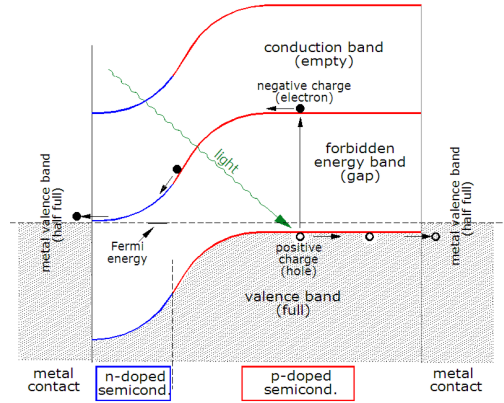


Figure 2. 3: Band diagram of a silicon solar cell, under illumination. Source: (Wikipedia, 2014)

The analysis of SC's functionality can be stated using two models: the first is related with a single diode model and the second to two diode model. However for the next steps we're revising the single diode model given that the second is more labour-intensive and is only used for full understanding of the behaviour of SC's (Haberlin, 2012).

In the single diode model there are two modes of functioning of the SC's, namely the ideal SC in which there are no crystal defects and, the second is related to the real SC in which crystal defects are taken into account.

a) The ideal SC is represented by a diode of three parameters: the short circuit current (I_{sc}), the diode current (I_d) and the current feeding the load (I).

In figure 2.4 below is presented a simplified diagram of an equivalent circuit of an ideal SC in which is taken into account that there is no voltage and current drop.

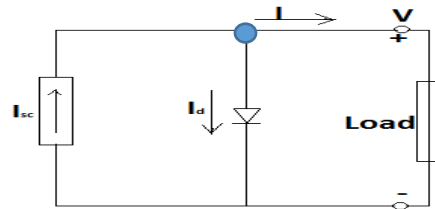


Figure 2. 4: Simplified diagram of an equivalent solar cell circuit

Applying the Kirchhoff's law to the blue node of the equivalent SC in the figure above, the short circuit current is given by:

$$I_{sc} = I + I_0 \left(e^{\left(\frac{qV_d}{mKT} \right)} - 1 \right) \quad (3)$$

Where I_0 is the diode reverse saturation current; V_d is the voltage across the junction; m is the diode ideality factor. The current flowing across the diode (I_d) is given by the term of (3), $I_0 \left(e^{\left(\frac{qV_d}{mKT}\right)} - 1 \right)$

There are two operating points of the solar cell, that can be obtained from the expression (3), in such a way that:

For short circuit ($V_d = 0$ leading to $I_d = 0$), thus:

$$I_{sc} = I \quad (4)$$

- For open circuit ($V_d = V_{oc}$, $I = 0$)

From equation (3) we get:

$$V_{oc} = \frac{mKT}{q} \ln \left(\frac{I_{sc}}{I_0} + 1 \right) \quad (5)$$

The open circuit voltage V_{oc} and short circuit current I_{sc} are parameters given by the manufacturer and are very important to draw the I-V curve, given below (Figure 2.5), which is used to predict the SC's performance at various temperatures, voltage loads and level of insolation (Yu, et al., 2010)

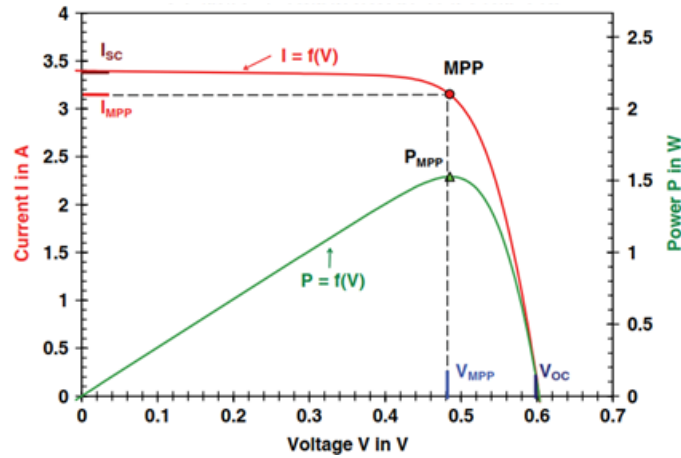


Figure 2. 5: I-V and P-V curves of a solar cell. Source: (Luque, et al., 2011)

The main important point in this curve is MPP (maximal power point) that is the point in which the module produces the greatest power, it is always found where the curve begins to bend and defines the outputs of the module (V_{MPP} – Voltage at maximum power point and I_{MPP} – Current at maximum power point). The power output at any other point is less than the power at maximum power point (P_{MPP}).

However, there is another parameter that describes the deviation of the I-V curve in relation to the ideal, called the Fill Factor (FF). The fill factor is the ratio of the maximum obtainable power to the product of the open-circuit voltage and short-circuit current, and is given by:

$$FF = \frac{I_{MPP} \cdot V_{MPP}}{I_{sc} \cdot V_{OC}} \quad (6)$$

The conversion efficiency (η) of SC, is the power density delivered at the operating point as a fraction of the incident light power density P_s and is given by the equation:

$$\eta = \frac{FF \cdot I_{MPP} \cdot V_{MPP}}{P_s} \quad (7)$$

b) The Real Solar Cell Model

The analysis of the real solar cell is made taking into account all cell defects namely, imperfection of semiconductor material defects, junction and metal contacts. These imperfections are clumped into Series Resistance (R_s) and Parallel resistance (R_p) shown in the figure 2.6 below.

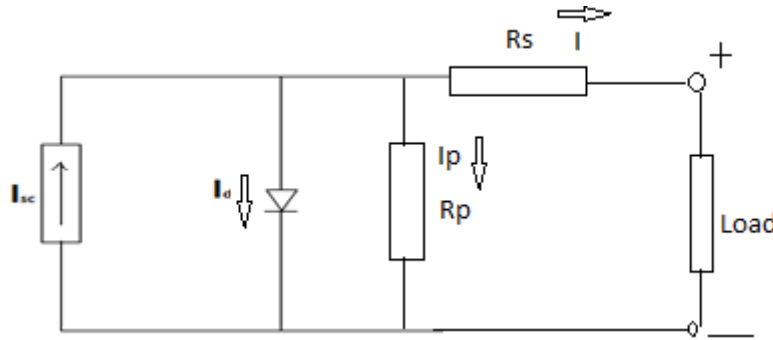


Figure 2. 6: Simplified equivalent real solar cell

Taking into account the figure above the short circuit current (I_{sc}) from equation number 3 and the current feeding the load can be transformed respectively to:

$$I = I_{sc} - I_0 \left(e^{\frac{q(V+IR_s)}{mKT}} - 1 \right) - \frac{V+IR_s}{R_p} \quad (8)$$

2.1.6 Factors that Influence the Operation of a Solar Module

There are five major factors that can affect the solar cell performance namely:

- a) **The cell material:** depending on the material and manufacturing method used, solar cells can achieve different conversion efficiencies of light, for instance the efficiency of amorphous silicon ranges from 5% to 7%, for the polycrystalline silicon, its efficiency does not exceed 12% and for the mono crystalline silicon the efficiency is over 12% and not exceeding 18% (Electronica, 2014) and (Seraphim, et al., 2004)
- b) **The radiation intensity on module output:** the current output is proportional to the radiation intensity, increasing the light intensity, the current also increases but the voltage does not change considerably at I-V curve above. (Haberlin, 2012).

c) **The heat on module output:** The output voltage of the module is affected by increasing the temperature of the module. At increasing temperature the voltage decreases considerably, however, the current does not change significantly. According to (Haberlin, 2012), the variation of temperature during the operation of a solar module may be explained by the equation below:

$$T_c - T_a = (NOCT - 20) \frac{G}{800} \quad (9)$$

Where G is the solar irradiance [W/m²], T_c and T_a are the cell and ambient temperatures respectively, “NOCT” is the Normal Operating Cell Temperature, its value is usually given by the manufacturer, and ranges between 44-54°C for a conventional module, under the following conditions at open circuit: G_{NOCT}=800W/m² of solar irradiance, T_a=20°C, air mass of 1.5 and Wind speed is 1 m/s.

d) **The shading effect on module output:** If a module or part of module (cell) is shaded, may partially produce or even not produce electricity, and if it happens can also cause hot spots heating. However, this fact can be mitigated installing bypass diodes (Stapleton, et al., 2012).

2.1.7 Solar Panel Orientation Angles

The optimal mounting position of solar panel varies according to the site latitude in north side of the equator the sun is usually in the southern position and the panel should be mounted facing South; and opposite for the south side of the equator. Taking into account that the sun moves in a 180° arc relative to the Ground from east to west, the solar altitude angle (γ) varies between 0-90° and the best absorption of solar radiation on a panel occurs when the striking angle is 90°, the energy yield can be increased by tilting the panel towards the Sun, i.e. at an angle β relative to the horizontal plane, generally for fixed panels this angle should be equal to the site latitude plus 10° (Figure 2.7) (Stapleton, et al., 2012).

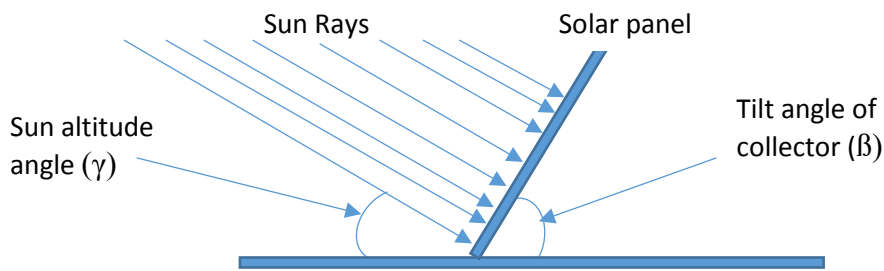


Figure 2. 7: The geometry of installation of solar panel

2.1.8 Solar panel circuit connections

The solar PV panels can be connected either in parallel or in series depending on the needs/purposes. Connecting in series, the voltage is increased while the current remains constant, while connecting in parallel would keep the voltage constant but the current is increased. However these connections can be combined both in one single circuit (series-parallel) as shown in figure 2.8.

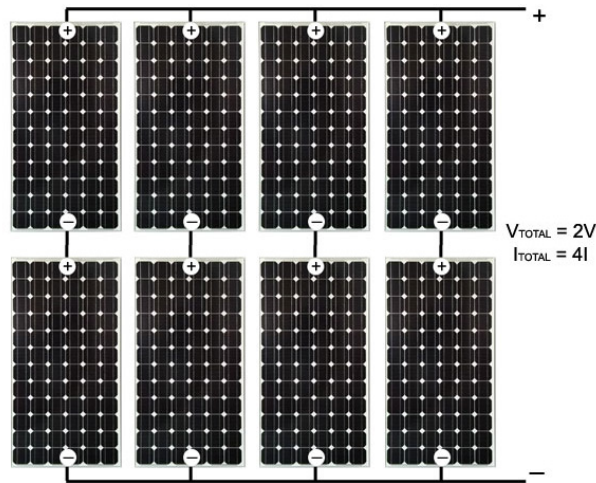


Figure 2.8: Connections of solar modules in series-parallel

The connections and results obtained from the solar panels are the same for Batteries connections.

2.2 The Wind Energy Resource

The wind is an abundant, free, clean, sustainable and environmentally-friendly renewable energy source. It has served the human civilization for many centuries by propelling ships and driving windmills to grind grain and pump water, and nowadays also for electrical power production (Johnson, 2006).

2.2.1 History of Wind Uses

Wind as source of energy has been in use by humans for many centuries and for energy purposes dates back five thousand (5000) years ago by sailing ships and boats used by the ancient Egyptians (Wortman, 1983), (Johnson, 2006). Its use occurs through the conversion of kinetic energy into a translational kinetic energy of rotation, using wind turbines to generate electricity, or windmills for mechanical work to pump water from deep wells (Burton, et al., 2011).

The first windmill machines were also used around 2000 years B.C., in ancient Babylon and China, evolving since then to the modern wind turbines. In the western world the first application of windmills was the grinding grain and pumping water from deep wells (Burton, et al., 2011), (IOWA, 2013) and (Coalition, 2013).

The first modern wind turbine designed especially for electricity generation, was constructed in Denmark in the early 1890 for supplying electricity to rural areas. In the same period a 12kW of power windmill was constructed in the United States of America (USA). It was a large wind electric generator with 17 meters diameter of rotor and 144 wooden propellers (Burton, et al., 2011).

The oil crisis in the 1970s sparked interest in wind energy in response to uncertainty of the price and availability of fossil fuels. The wind turbine technology research and development programs that followed this oil crisis during last century, introduced significant improvements resulting in modern computer

controlled wind turbines, which brought new and more efficient ways of converting wind energy into useful mechanical or electrical power (Coalition, 2013), (Ahmed, 2011).

Nowadays the mankind is looking for new sources and technologies of energy that are inexpensive, regarding the exhaustion of coal, oil and even radioactive materials. For this reason there is a multitude of scientific studies tending to better leverage inexhaustible sources of energy such as the renewables and the solar and wind in particular.

2.2.2 Wind Turbines: working principles

Most wind turbines (WT) are machines built to convert the containing power in the wind into electricity. The main classification of those machines is according to the interaction of their blades with the wind by aerodynamic forces - drag or lift or a combination of both; and the orientation of the rotor axis with respect to the ground and to the tower – upwind or downwind (Ahmed, 2011). According to the orientation of the axis there are two types: The Horizontal Axis Wind Turbine, or HAWTS, and Vertical Axis Wind Turbines, or VAWTS (Figure 2.10).

Among the VAWT's machines we highlight the Savonius (Figure 2.10 b) mostly used for water pumping and the Darrieus (Figure 2.10 a) WT. They have the advantage of receiving wind from any direction not requiring tracking mechanisms of the wind direction and that the coupling between the rotor and the generator can be made at ground level, allowing easy access for maintenance meaning that smaller towers gets reduced costs. The main disadvantage is that it has no self-starting, high torque fluctuations and limited options of regulations at high wind speed.

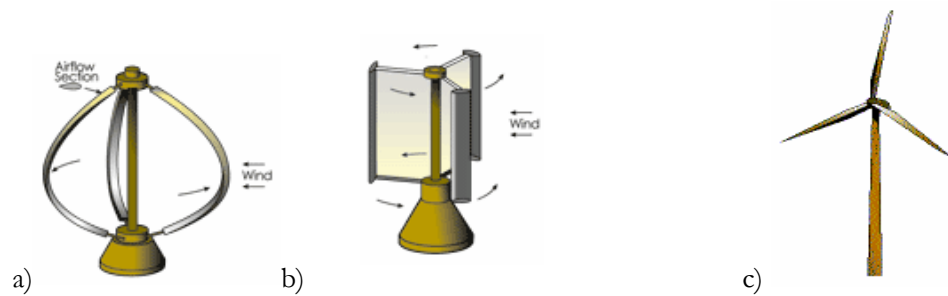


Figure 2. 9: Types of rotors according to the orientation of its axis. (a) Darrieus rotor - Vertical Axis (b) H-type rotor - vertical axis, and (C) Upwind horizontal axis rotor. Source: (Burton, et al., 2011) and (IOWA, 2013),

The other type is the HAWT's where the rotors are kept perpendicular to the wind and the rotational driving force is lift and the blades can be in front (upwind) or behind (downwind) of the tower. The HAWT's take advantage of extracting higher wind speeds farther from the ground as the rotors are placed on the top of a tower.

Detailed explanation of the working mechanisms of both WT's types can be found in the literature such as (Manwell, et al., 2009), (Ackermann, et al., 2000), (Ahmed, 2011), (Burton, et al., 2011), etc.

In the present work we will focus on the HAWT's with three blades attached to a central hub as it is the most widespread in the wind power industry and in use today. Together, the blades and the hub form the rotor (the main element to capture energy), which are connected to an electrical generator. When the wind blows, the rotor turns and the generator produces alternating current (AC) electricity. WT with multi-blade rotors (20 or more blades) have high starting torque in light wind and are mainly used for mechanical water pumping.

The main configuration and components of the HAWT are shown in figure 2.11, which consists of a tower and nacelle mounted at the top of a tower.

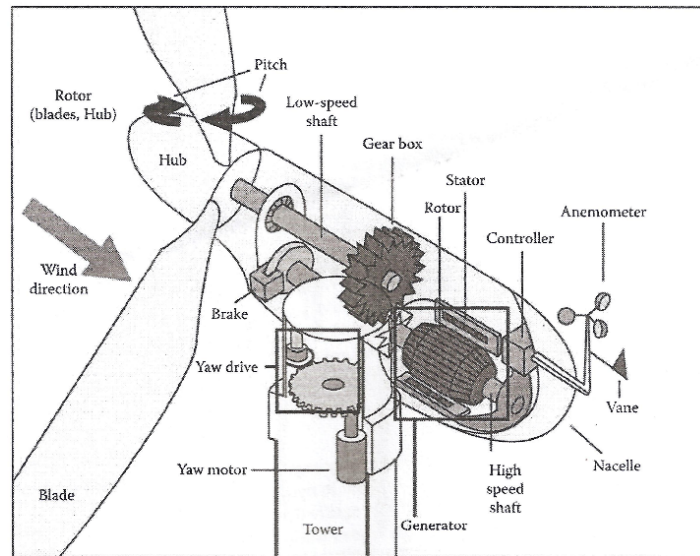


Figure 2. 10: Diagram of parts that constitute a wind turbine. Source: (Burton, et al., 2011).

The nacelle contains the main components of the WT such as: the electricity generator, gearbox and the rotor. The generator transforms the rotational mechanical energy delivered by the gears, into electrical energy. The generator may be of asynchronous, synchronous, direct current and alternating current commutated types each having its advantages and disadvantages, the use of one type or the other will depend on the turbine size or the specific application and on the preferences of the manufacturer of the turbine (IOWA, 2013), (Burton, et al., 2011).

2.2.3 Power Output Control

Most of WT machines have control systems, composed of wind speed and direction measuring devices (anemometers) and computer systems connected to sensors, valves, motors and pumps that monitor all processes and trigger mechanisms across the turbine.

These control systems communicate with the operator via a communication link sending alarms or requesting for services by means of radio, telephone and nowadays through the Internet. It also can gather statistics or check the status of wind turbines.

The main mechanism to point here is yaw control that orients the turbine (points the nacelle) towards wind direction or moves the nacelle out of the wind in case of high wind speeds in response to a signal from the wind. For small WT, the rotor and the nacelle are oriented into the wind with a tail vane (Ahmed, 2011).

There are other three principles of aerodynamic control, aiming to maximize the extraction of power from wind turbines which are: The Passive Stall and, active Pitch Control, and active stall.

The **passive stall control** is a system which reacts according to the wind speed. In this system the blades are fixed according to a given pitch angle so that there is a decrease in the aerodynamic forces of lift coefficient and associated increase in drag coefficient when wind speeds above the rated speed are achieved (Ahmed, 2011).

The **active pitch control** is a system requiring information from the control system via an anemometer or other sensors installed in the turbine. The pitch control operates when the rated power of the turbine (cutting speed) are exceeded, the rotor blades rotating around its axis, reducing the angle of attack of the wind and thus the aerodynamic forces on the turbine, reducing the extracted power. The rotor blades are rotated at certain angles, for each wind speed above to the rated power to continuously extract the rated power (Ahmed, 2011).

The **active stall control** combines both pitch and stall control mechanisms. This type of control achieves power limitations above the rated wind speed, by pitching the blades initially into stall, i.e. there is a small turning of the blades (typically up to 5°) around its axis for certain speeds. The twists along the blades are necessary with this type of control (Ahmed, 2011).

Three other factors on the turbines are important: the starting speed, the rated/nominal speed and cutting speed where starting speed is the minimum wind speed needed to start moving the blades and start producing some energy (Ahmed, 2011). The rated speed is the minimum speed at which the turbine is designed to develop the rated power

And finally the wind turbines also have an aerodynamic braking systems provided on each blade so that in strong winds the rotor can be cut off. Each WT comes with information on the cut-off speed, that is the maximum speed at which the turbine ceases or decreases energy production, activating aerodynamic control systems or brakes, avoiding damages into the turbine structure or in the electric distribution system (IOWA, 2013), (Burton, et al., 2011).

2.2.4 Wind Power Generation Technology

The wind is considered as a vector defined by: the wind direction and wind speed. Wind direction is the direction from which the wind blows and is expressed in degrees. The wind speed is expressed in meters per second (m/s), kilometres per hour (km/h). Measure of wind strengths over a period of time and in different directions can be well illustrated and analysed through a wind rose diagram. This diagram can indicate the percentage of time for which we receive wind, speed and energy from a particular direction. A typical wind rose is a circular display of how wind speed and direction are distributed at a given location for a certain time period.

The figure 2.11 below illustrates the distribution of three variables in the wind rose diagram displaying values which are related to compass direction. The plots represent the percentage of time distribution of one year for which wind blows from that direction.

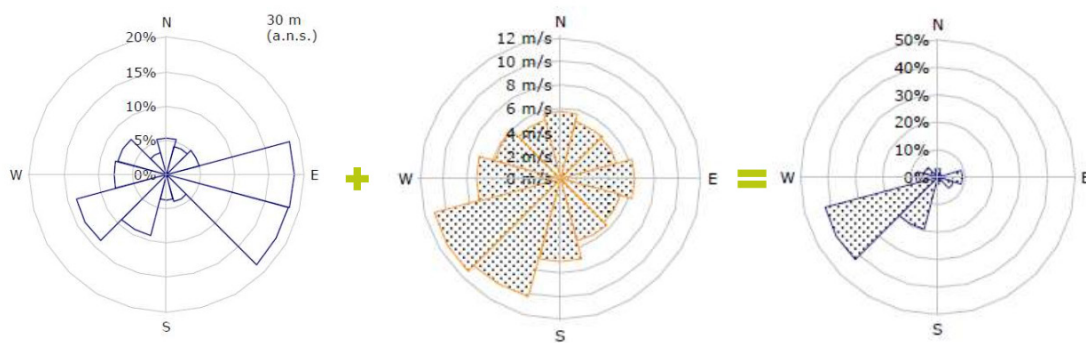


Figure 2. 11: (a) Direction per sector (b) average speed by sector (c) energy per sector

The wind direction and its speed are very important to harness the wind power, since frequent changes in wind direction indicate gust conditions. Wind data from the meteorological services are very useful, however, in most cases the data is not available. Thus available data from the nearby meteorological stations can be used to give us an indication on the wind spectra available at a specific site (Ahmed, 2011).

2.2.5 Wind Power Conversion

To measure the wind potential of a given site is considered a particular wind flow passing through a cylinder with a cross *section area* A , given by the diameter of a rotor (Figure 2.12).

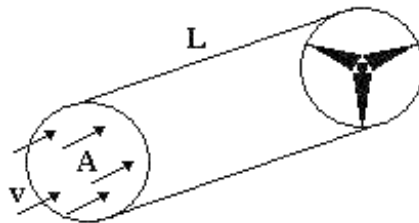


Figure 2. 12: Flow of wind through a cylinder/rotor of area (A) and length. Source: (Rohatgi, et al., 1994).

According to the laws of physics, the rate of kinetic energy $E = P$ of the air mass flow m with a Speed V will be:

$$P = \frac{1}{2} \dot{m} V^2 = \frac{1}{2} (\rho A V) V^2 \quad (10)$$

Where $\dot{m}$ is the mass flow rate of the air, ρ the air density (1.225 kg/m³ at sea level) that depends on altitude and meteorological conditions - air pressure and temperature, both being functions of height above sea level. (Walker, et al., 1997; Gipe, 2004) (Manwell, et al., 2009). The expression 1 represents the total power which is given in Watts (W).

Considering a wind turbine placed inside of the cylinder, and part of the wind power will be transferred and used by this wind turbine, then the power output “P” extracted by the rotor will be

$$P = \frac{1}{2} C_p \rho A V^3 \quad (12)$$

Where C_p is a constant, dimensionless power coefficient or Betz limit, and is a measure of the efficiency of the wind turbine in extracting the kinetic energy content of a wind stream that may be converted into mechanical work, A is the rotor swept area. (Wortman, 1983); (Walker, et al., 1997) and (Twidell, et al., 2006). It has a theoretical maximum value of 0.593, i.e., $C_p = \frac{16}{27} = 0.593 = 59\%$.

On both expressions we can perceive that any increase in wind speed, will result in a substantial increase of the power contained in the wind flow, but according to (Rohatgi, et al., 1994), only a part of the wind power can be converted into a useful power, which in turn the available power from the wind turbine is not limited only by the Betz coefficient but also on the aerodynamic and mechanical losses in the turbine those related to rotation at the tip and base of the blade and the effect of the number of blades, as well as on the limitation of the chosen rated generator capacity. The efficiency of mechanical equipment such as the multiplier gearbox, the electrical generator and its coupling have to be considered.

Another important concept relating to power coefficient of wind turbine rotor is the optimal tip speed ratio (TSR, λ_0), that is, *the ratio of the rotor tip speed to free wind speed*. Rotor performance can be optimum only for a unique tip speed ratio. The TSR of the rotor depends on the characteristics of the used blade airfoil profile-blade radius “R”, oncoming wind speed V_0 , the angular velocity ω , the number of blades “n” and the type of wind turbine (Twidell, et al., 2006).

In general, a three bladed wind turbine operates at a TSR between 6 and 8, with 7 being the most widely reported value (Rageb, et al., 2011), (Twidell, et al., 2006) and (Ackermann, et al., 2000) thus the TSR is dimensionless factor defined as:

$$TSR = \lambda_0 = \frac{\text{speed of rotor tip}}{\text{wind speed}} = \frac{R}{V_0} = \frac{\omega R}{V_0} \quad (13)$$

$\frac{\omega R}{V_0} \approx \frac{4\pi}{n}$, Where, V_0 is the speed (m/sec) of the oncoming wind, ϑ is the rotor tip speed (m/sec), R is the radius (m), $\omega = 2\pi f$ is the angular velocity (rad/sec) and f is the rotational frequency (Hz), (1/sec).

2.2.6 Prediction Models For Wind Resource

The wind resource is very variable in nature. Combining meteorological and statistical techniques to forecast wind can give us a very useful predictions for a specific wind farm power output project in which will give us a best idea to choose the appropriate Wind Energy Conversion System (WECS).

In the wind industry the Weibull probability distribution function (PDF) is commonly used. The Weibull and Rayleigh PDFs are the most widely used PDFs to describe the wind speed distribution for WECS applications (Ahmed, 2011) and (Twidell, et al., 2006).

According to (Twidell, et al., 2006), the wind speed V is distributed as the Weibull distribution, being the probability distribution function expressed by

$$F(V) = \frac{8760}{c} k \left(\frac{V}{c}\right)^{k-1} \exp\left[-\left(\frac{V}{c}\right)^k\right] \quad (14)$$

($k > 0, V > 0, c > 1$)

The Weibull distribution is a two parameter (c and k) function while the Rayleigh distribution is a one parameter (c) as the Rayleigh distribution is a special case of the Weibull distribution in which the shape factor k is 2 (Spera, 2009).

In the equation above $F(V)$ is the PDF that expresses the frequency of occurrence of a certain wind speed in (m/s) during a year, $\exp$ is the base “e” exponential function, c (m/s) and k are the scale parameter and shape parameter (dimensionless) respectively.

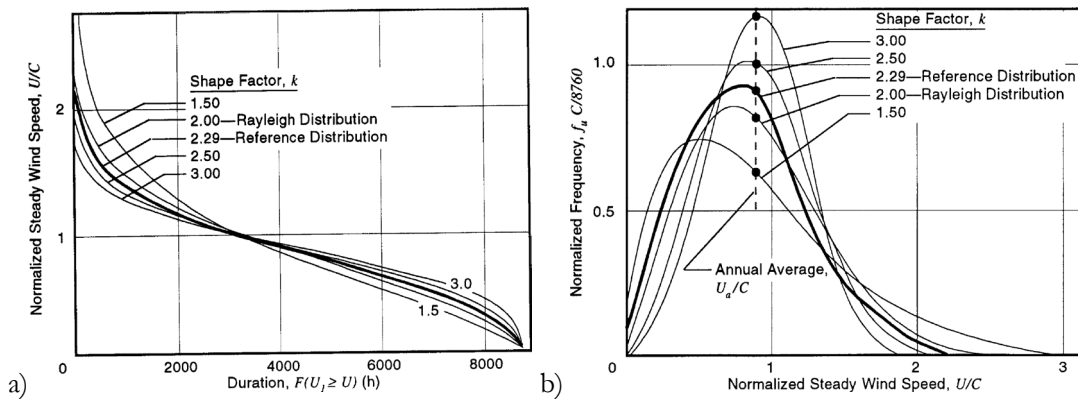


Figure 2.13: a) Wind speed duration curves according to the Weibull distribution model. b) Wind speed frequency distribution. Source: (Spera, 2009)

The figure 2.13 (a) shows the Weibull frequency distribution $f(V)$ curves associated with the duration curves and in figure 2.13 (b) the annual average wind speed for each value of k . It is noted that the

Weibull density function gets relatively narrower and more pinched as **k** gets larger. The peak also moves in the direction of higher wind speeds as **k** increases.

According to (Manwell, et al., 2009), another important indication for wind energy resource evaluation of a specific location's wind energy characteristic is the mean **wind power density (WPD)**, defined as the wind power available per unit of swept area by the turbine blades and is tabulated for different heights above ground and given by the following expression:

$$WPD = \left(\frac{P}{A} \right) = \frac{\rho V^3}{2} \text{ or } WPD = \frac{1}{2n} \sum_{i=1}^n \rho V_i^3 \quad (15)$$

Where **n** is the number of records in the averaging interval, **ρ** air density and **V_i³** the cube of the **ith** wind speed value in **W/m²**

Changes in air velocity and its density affect the WPD. The table below shows the classes of wind power density at 10m and 50 m²

Table 2. 1: Classes of wind power speed and density mean values ². Source: (Ahmed, 2011)

Wind power class	Description	10m		50m	
		Wind power density (Watts/m ²)	Wind Speed (m/s)	Wind power density (Watts/m ²)	Wind Speed (m/s)
1	Poor	<100	<4.4	<200	<5.6
2	Marginal	100 - 150	4.4 - 5.1	200 - 300	5.6 - 6.4
3	Fair	150 - 200	5.1 - 5.6	300 - 400	6.4 - 7.0
4	Good	200 - 250	5.6 - 6.0	400 - 500	7.0 - 7.5
5	Excellent	250 - 300	6.0 - 6.4	500 - 600	7.5 - 8.0
6	Outstanding	300 - 400	6.4 - 7.0	600 - 800	8.0 - 11.9
7	Superb	>400	>7.0	>800	>11.9

Areas designated class 3 or greater are suitable for most utility-scale wind turbine applications, whereas class 2 areas are marginal for utility-scale applications but may be suitable for rural applications. Class 1 areas are generally not suitable, although a few locations (e.g., exposed hilltops not shown on the maps) with adequate wind resource for wind turbine applications may exist in some class 1 areas. The degree of certainty with which the wind power class can be specified depends on three factors: the abundance and quality of wind data; the complexity of the terrain; and the geographical variability of the resource. Vertical extrapolation of wind speed is based on the 1/7 power law (NREL, 2014).

²⁾ Vertical extrapolation of wind speed based on the 1/7 power law and Mean wind speed is based on the Rayleigh speed distribution of equivalent wind power density

2.2.7 Factors that Affect the Wind Characteristics (Speed and Power)

Boundary Layer Effects

The wind speed and its available power are very influenced by regional and local factors. The most important are the variations in wind velocity due to the boundary layer/surface effect. The earth surface offers a frictional resistance (due the roughness of the ground, vegetation, etc,) to the wind flow. The wind velocity varies (increasing) with the height above the ground (Mathew, et al., 2011), and this phenomenon is also called wind shear. (Manwell, et al., 2009).

Figure 2.14 shows an example of wind profile at a specific site, where the variations in the wind velocity with height can be observed.

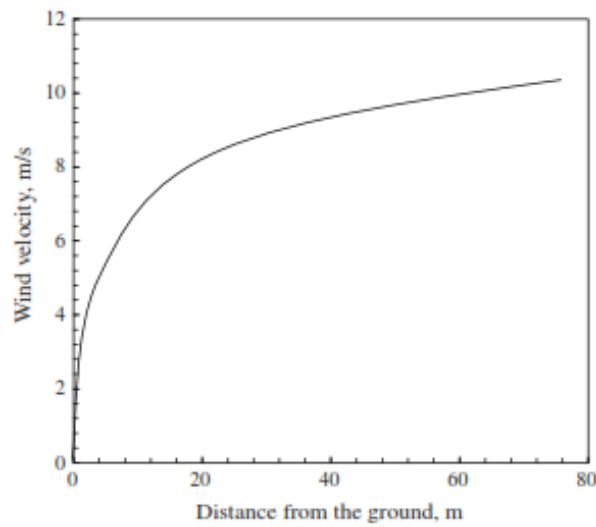


Figure 2. 14: Typical variation of wind velocity with height above the ground.

In some wind energy projects the wind data is collected from the hub height of the turbine, and if the available data were not collected from the hub height it is necessary to correct it for the boundary layer effect. At this stage the ground resistance against the wind flow takes place here represented with the roughness class or the roughness height (Z) of a surface that may be close to zero (surface of the sea) or even as high as 2 (town centers).

According to (Walker, et al., 1997) the logarithmic profile (log law) is commonly used to describe the vertical variation of mean wind speed with height within the lowest portion of the boundary layer (surface layer).

The logarithmic wind profile, equation (16) below, is used for the estimation of the mean wind speed (V) at height (Z):

$$V(z) = \frac{V^*}{k} \ln \left(\frac{z}{z_0} \right) \quad (16)$$

Where V^* is the friction velocity, k is the Von Karman's constant that is equal to 0.4 and $V(z)$ is the mean wind speed at height Z and Z_0 is the surface roughness length which characterizes the roughness of the ground terrain.

Some typical values of approximate surface roughness lengths are given in Table 2.2.

Table 2. 2: Typical values of surface roughness length Z_0 of the terrain. Source: (Mathew, et al., 2011)

Terrain Category	Class	Surface	Landscape Description	Z_0 (m)
1	1	Sea	Open sea, fetch at least 5 km	0.0002-0.005
1	2	Smooth	Mud flats, snow, little vegetation, no obstacles	0.005
2	3	Open	Flat terrain: grass, few isolated obstacles	0.025-0.1
3	4	Roughly Open	Low crops: occasional large obstacles	0.1
3	5	Rough	High Crops: scattered obstacles	0.2-0.3
3	6	Very Rough	Orchards, bushes: numerous obstacles	0.5-1
4	7		Tall Forests, Dense Town Centers, etc.	1-2

The log law equation can be modified so that can be used to extrapolate from a reference height Z_r to another level (to the hub height of the turbine for example) using the relationship:

$$\frac{V(z)}{V(Z_r)} = \ln\left(\frac{Z}{Z_0}\right) / \ln\left(\frac{Z_r}{Z_0}\right)^\alpha \quad (17)$$

Where $V(Z_r)$ and $V(z)$ are the wind speeds at reference height (eg. $Z_r = 10m$) and at the new height (Z) respectively and Z_0 is the roughness length. Another approach that is used by many researchers for a simple estimative of the distributions of mean wind speed with height is the empirical model: power law (Walker, et al., 1997), given by:

$$V(z) = V(z_r) \left(\frac{z}{z_r}\right)^\alpha \quad (18)$$

where $V(z)$ is the needed wind velocity at elevation Z , $V(z_r)$ is the available mean wind velocity at the known higher elevation Z_r ($=10m$) and α is the terrain roughness exponent and is also known as empirical wind shear/power law exponent that is generally obtained experimentally from measurements of wind speed at different heights and can be determined as:

$$\alpha = \frac{\log \frac{V(z_r)}{V}}{\log \frac{z_r}{z} - 1} \quad (19)$$

In assessing the potential energy of a wind turbine it is very important to convert or extrapolate the available wind data to the height of the turbine rotor shaft. According to vertical extrapolation of wind speed and under certain conditions α is equal to 1/7, indicating a correspondence between wind profiles and flow over flat terrain.

Table 2. 3: Typical values of the power law exponent α . Source: (Mathew, et al., 2011)

Terrain Category	Class	Type of terrain	Landscape Description	α
1	1	Sand		0.1
1	2	Mown grass		0.13
2	3	High grass		0.19
3	4	Suburb buildings		0.32

Two methods for modelling the vertical wind profile (the log and power laws) were discussed in this section. Those laws are applicable for flat and homogenous terrains meaning that if we consider irregular earth surfaces it is expected to have modifications to the wind flow mainly if we're dealing with flat and non-flat terrains. More detailed explanation of their effects to the wind flow can be found in literature.

2.3 Energy Storage (Battery types and operation)

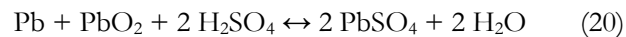
One of the problems that energy production from renewable energy source sector faces, is to maintain fixed or constant the amount of electricity produced over a certain period of time, even though knowing that throughout the day electricity production and demand fluctuate. To overcome this problem, energy storage systems are used, which capture excess energy during periods of low demand storing it in other forms and convert it back when needed to feed the demand.

In the case of power generated from wind and solar sources, rechargeable batteries, also called accumulators are used. Batteries are electrochemical devices used to store electric energy by converting it into electrical charges in the form of ions. In case of electrification in remote areas using solar or wind energy, the optimum battery would cover the demand during the night, on cloudy and rainy days or at lower wind speeds. These devices are also important to stabilize the large voltage fluctuation produced by a solar module and wind turbine (Hankins, 2010).

2.3.1 Types of Batteries

The rechargeable batteries can be either lead-acid, nickel-cadmium, nickel metal hydride, or lithium ion type for the ones being used commercially and in small scales. The lead acid batteries are most common for solar/wind power systems due to their suitability, availability and low cost compared to the other types which are most used for small electric appliances such as radios and cell phones (SEI, 2004).

Lead Acid Batteries: the basic principle of operation of lead-acid batteries is based on the reaction of lead plates coated with PbO_2 (negative plates) that are connected to the positive connector (Pb) while the lead plates (positive plates) are connected to the negative connector. They are separated by a cardboard, plastic or some micro porous paper separator and then the assembly is placed in the battery compartment and dipped in an aqueous solution of sulphuric acid (H_2SO_2) as shown in the reaction equation below (Hankins, 2010).



According to Hankins, 2010, the lead-acid batteries can be classified into two categories: the automotive battery (start-up battery) and the deep discharge (deep cycle) battery.

Automotive batteries are designed to provide high current peaks for short periods, resulting in a small depth of discharge which is usually only 20% of the charge capability. This type of batteries are mostly used for engine starting, given that at the time of starting the starter of a vehicle's engine consumes a lot

of power for a short time. Batteries designed for peak current differ from the stationary deep-cycle ones by having more plates, but thinner (SEI, 2004).

Deep discharge (deep cycle) batteries are designed to withstand discharges of up to 80% of their capacity, for instance absorbed glass mat (AGM), captive electrolyte gel and tubular plate batteries/OPZS or OPZV (wet or gel Cells) batteries.



Figure 2. 15: Different types of deep discharge batteries, AGM (a); captive electrolyte gel (b) and tubular plate batteries/OPZS or OPZV (c). Source: (PALSOLAR, 2014)

There are differences between the different deep-cycle battery modifications regarding architecture, prices in the market and number of life cycles; however, the most preferable is the last one in Figure 2.15 – the Tubular Plate type – due to its exceptionally long lifetime (900 to 1200 cycles) compared to any other lead-acid battery type (SEI, 2004).

2.3.2 Key Battery Parameters and Characteristics

a) Battery Voltage (V)

The nominal voltage of a lead-acid battery is by definition **2.0 V** per cell; however, this voltage varies largely during charging and discharging, as a function of the current delivered or withdrawn, the elapsed time of loading or unloading, the temperature and constructive characteristics. During fast charging or if the battery is overaged, the voltage may reach 2.5 V per cell. During deep discharge, the voltage may drop to 1.6 V per cell, which is commonly regarded as a destructively low level.

b) Battery capacity (C)

The battery capacity is usually defined in ampere-hours (Ah) and it is the amount of electricity that the battery is capable of providing under certain conditions, i.e., with given discharge current until a certain voltage level at a certain temperature. The battery capacity and discharge current are often indicated in conjunction with a subscript for discharge time in hours, i.e., C10 meaning battery capacity C for a discharge time of 10 h, is normally used to designate a battery's nominal capacity and it is used as base for

comparisons of different battery capacity data according to approximation below as C20 and C100 battery are the most useful for stand-alone systems.

Approximate expression for nominal capacity:

$$C_{10}: C_{10} \sim 0.85 \times C_{20} \sim 0.7 \times C_{100} \quad (21)$$

c) Battery charge and discharge limit

If the battery is connected to the load, its voltage starts to decrease however if its nominal voltage becomes around 1.7 to 1.85 V, the battery must be disconnected from appliances in order to prevent a destructive depth of discharge (DoD) and enable long life span. If it is being charged and the voltage becomes higher than around 2.4 V, the battery has to be disconnected to prevent elevated gas formation (gassing)³ (Haberlin, 2012).

d) Life cycle versus depth of discharge

Life cycle is the number of cycles that a battery can carry out before its capacity reaches 80% of nominal capacity and it is basically determined by battery type and DoD. The higher the DoD, the shorter the life span for all types of batteries. It is advised that even deep cycle batteries should not be regularly discharged below 60% of DoD (40% state of charge) (Haberlin, 2012).

Moreover, all types of batteries perform better at low currents rather than at high ones, both for charge and discharge. Slow charge/discharge procedures prolong the life span of any battery and allow for a sustained high capacity level throughout the life cycle. Fast charging as well as fast discharge drawing high currents can easily lead to worse performance and shorter lifetime for any electrochemical battery type.

2.4 Inverters and Charge Controllers

Inverters are devices used to convert the DC electrical current into AC and vice versa. The inverters can be classified according to their waveform: square wave, modified square wave and sine wave (SEI, 2004).

The **square wave inverters** have a very weak control voltage output and a high harmonic distortion, hence they are not suitable for residential use in spite of having the lowest costs on the market. The **modified square wave inverters**, are coupled with some electronic devices (field effect transistors – FET or Silicon controlled rectifiers -SCR) which enable them to reduce the harmonic distortion presented by the first type. However, certain residential equipment cannot be connected to this type of inverters either. **Sine Wave Inverters** - are the most advanced and appropriate for residential electrification purposes, since they provide an output signal with low harmonic distortion. That fact allows them to supply electricity even to any equipment, including sensitive devices (SEI, 2004).

³⁾ Gassing is the process of destructive gas generation in batteries when they are overcharged

Charge controllers are electrical circuits used to control the state of charge of the batteries. These devices work as an enter key, allowing the current to flow into the batteries when it is charged and close the current flow when the batteries are fully charged in order to prevent overcharge as referred in the section 2.3.2 c) above. In most cases the controllers are coupled into inverters (Hankins, 2010).

2.5 Evaluation of Solar Energy and Wind Energy Resources

Solar or Wind power generation does not supply electricity to the load continuously, due to its intermittent character preventing it from meeting a steady constant demand at different times. Therefore, both sources need to be considered as variable forms of energy output. Their separate utilization should always account for the variability and unpredictability of the resource. (Gipe, 2004).

One way to minimize the influence of intermittency of wind and solar sources is to combine the two sources into one system so that the unavailability of one of them can be compensated by the activity of another. This combination leads to hybrid configurations, which are the focus of the present study.

2.5.1 Electricity Supply Through Hybrid Power Systems

Hybrid power systems are basically those systems that consist of two or more energy sources for power generation, and can be conventional or not such as generation by wind, solar PV, natural gas, diesel oil, biofuel among others, with the goal of providing electricity or cogeneration, either stand-alone or connected to the grid. Those systems are complex and require the optimization of energy control and use of all sources in order to get the maximum efficiency on the delivery of energy to the consumer units, maintaining the quality and reliability specified for each proposed project.

Within this present study, an isolated solar-wind hybrid system for electricity generation and a mini-grid to distribute the generated electricity to a small load are proposed. Figure 2.16 illustrates the schematic example of a hybrid system configuration of this type.

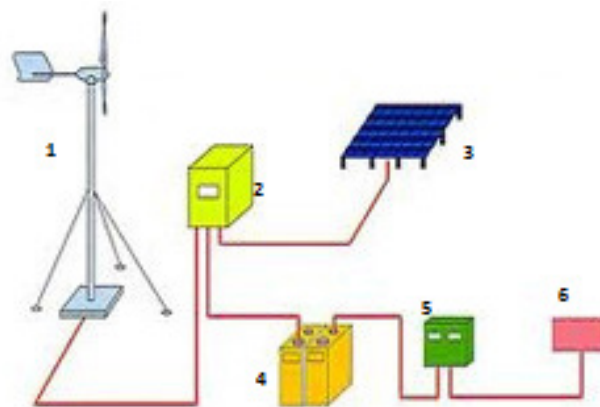


Figure 2. 16: Stand-alone hybrid system schematic configuration with AC energy bus: 1- Wind turbine; 2- Charge controller/bidirectional inverter; 3- Solar PV panels; 4- Battery bank, 5- Grid Inverter; 6- AC appliances. Source: (BCS, 2014), modified by Silinto & Bila, 2014.

The system combines solar photovoltaic panels that produce electricity in DC and wind turbines, which can produce electricity either in DC or AC. To associate these sources of electricity to a load and battery bank, whose current is continuous (DC), the system needs a power converter that can ensure that current flows between the different equipment and provides to the load indicated by “6” in the figure above.

2.5.2 Advantages of Hybrid Power Systems

Hybrid systems that use renewable energy sources, such as solar and wind resource, may be feasible and an alternative to supply electricity to remote or isolated areas from the national grid and help in reducing the use of fossil fuels, dependence on costly fuel, and reduce the emission of greenhouse gases.

The main advantages of hybrid systems compared to single ones are presented below:

- **Complementarity between sources of the system:** intermittency of sources involved can be partially or completely overcome, ensuring continuity and quality of the electricity produced by the system;
- **Modularity of the involved sources:** photovoltaic modules, turbines, and batteries can be purchased gradually of the system, provided there is natural growth of the system in line with the availability of financial resources, energy potential and area for the system installation;
- **The socio-economic impacts**, in general, are characterized as products for the deployment of hybrid systems. In most instances the impacts have more beneficial features than harmful, especially when treating with small generation systems as is proposed in the present work. It is also noted the Rural population growth, because it can combat the rural exodus, the increase of local commercial activities (bars, warehouses), agriculture, education and telecommunication, which is made possible by the wider use of electricity;
- **As for environmental impacts**, they are primarily related to the end-of-life issues and recycling of the used equipment, primarily the management of batteries (there is a need for a recycling program); to the visual and noise aspect from the rotation of the wind turbines blades, and occupation of space used for equipment installation of the hybrid system. Hybrid systems may represent a viable alternative for technical, financial, social and environmental criteria, including advantages with regard to the extension of the power grid or the local power generation by diesel systems.

2.6 The Local Study Area (Mozambique)

2.6.1 Geographical Location and Climate

Mozambique is located on the Eastern African coast side of the southern African subcontinent, facing the Indian Ocean between latitudes 10° 27'S and 26° 52'S and between longitudes 30° 12'E and

40° 51'E, and has a long coastline of about 2,515 km with continental shelf area of approximately 68,000 km² (INE, 2007). Figure 2.17 illustrates the Map of Mozambique and its regional division.

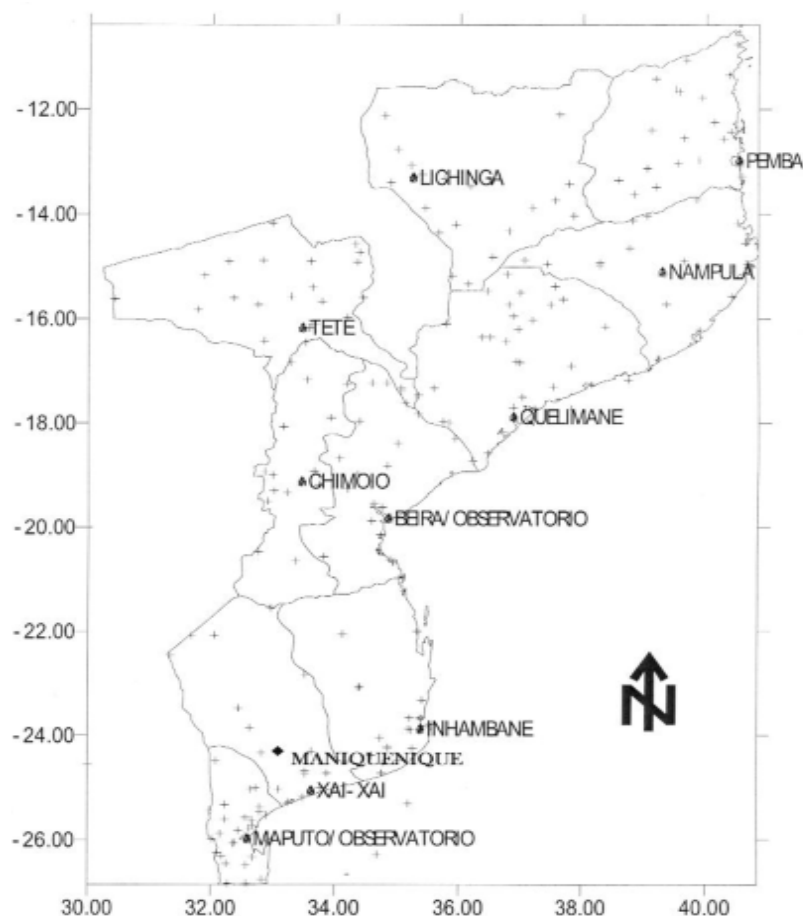


Figure 2. 17: Map of Mozambique. Source: (Cuamba, et al., 2006)

Its climate is characterized by two typical climate seasons: the wet season, which extends from October to April and is hot and rainy; and the dry season, which is a kind of a cooler winter from May to October with very limited rain. The annual mean temperature vary from the northern to the southern region and from the coast to the highlands in the interior of the country. The coastal plain has a tropical humid to sub humid climate with a mean temperature of about 24°C in the southern region and 25°C in the northern region (Zucule, 2012)

2.7 Mozambique Energy Profile

Mozambique is rich in energy resources (fossil and renewables) not exploited yet: namely natural gas and coal, solar, wind, hydro, geothermal, ocean and biomass. According to Hankins, 2009, it is estimated that the renewable sources are as large as presented in the Table 2.4 below:

Table 2. 4: Renewable Energy Resources in Mozambique

Resource	Availability	Comments
Biomass / Cogeneration	100s of MW from various sources potential bagasse availability of 433 thousand ton (dry weight, 2006)	5 sugar plantations located in Maputo and Sofala Provinces
Wind	Encouraging wind resources exist along the coast and in Niassa Province Tests shows >6m/s average wind speed in some areas	Site studies using 10 and 20 meters masts will probably reveal greater resources. Updated resource mapping needed.
Solar	High 4.5-7 kWh/m ² /day Assuming average insolation of 5.2 kWh/m ² /day, 1.49 million GWh of annual radiation is incident on Mozambique land surface	Estimated 1 MW of off-grid PV systems installed. FUNAE study on PV potential under way.
Small-scale hydro (up to 10 MW)	>1000 MW	60 potential projects
Geothermal	Possible resources, but no studies completed yet Conservative estimates of at least 25 MW in Tete Manica and Niassa provinces	No realistic plans or resource assessments
Tidal & Wave	Ample resources, but no studies completed yet	No realistic plans or resource assessments

Source: (Hankins, 2009), adapted by Silinto & Bila 2014.

The available mapping of the distribution and potential of the solar and wind resources in the country are presented in the next section.

2.7.1 Solar Radiation and Wind Regime, Distribution and its Potential

a) Solar Radiation Resource:

Mozambique has great unexploited solar radiation potential. According to Cuamba (2006), the availability of solar radiation across the country, varies between 5.2 and 6 kWh/m²/day with the annual daily average of 5.7 kWh/m²/day. However the most recent studies done by FUNAE, confirm that in Mozambique the global horizontal irradiation varies between 1.785 and 2.206 kWh/m²/year and the solar potential is 23 TWp (FUNAE, 2014). Figure 2.18 below shows the solar radiation distribution in Mozambique.

b) Wind Resource:

In general, coastal areas of Mozambique are characterized by more frequent strong winds than interior areas. This contrast is related to the thermal effect and the low roughness of the water surface in coastal areas where is relatively stable throughout the year with higher intensity between the months of September to November.

Figure 2.19 below shows the annual wind distribution in Mozambique, where the highest wind energy potential is observed in the provinces of Maputo, Tete, and Sofala, Inhambane and coastal area of Gaza province.

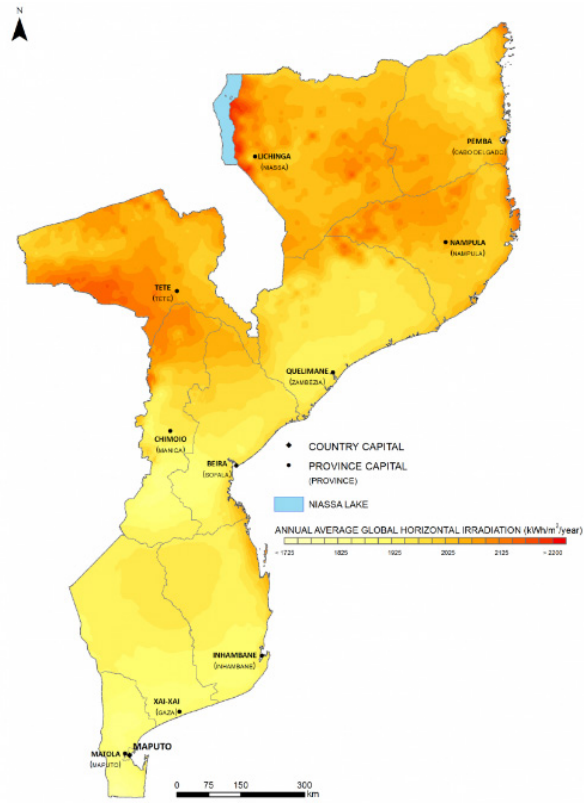


Figure 2. 18: Solar radiation distribution for Mozambique. Source: (FUNAE, 2014)

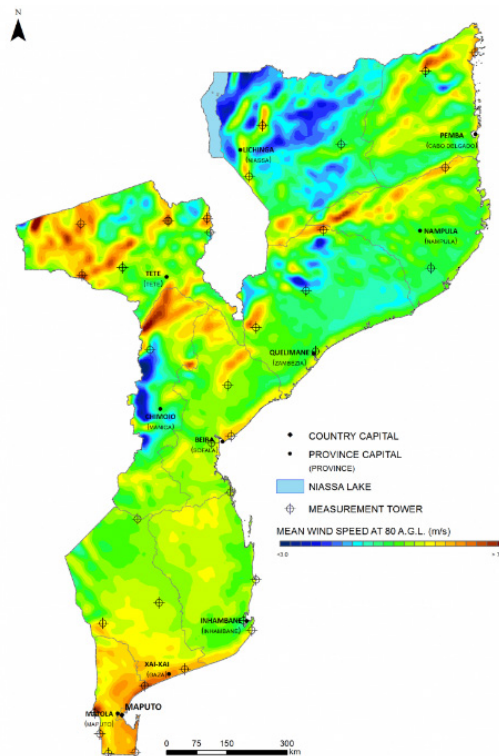


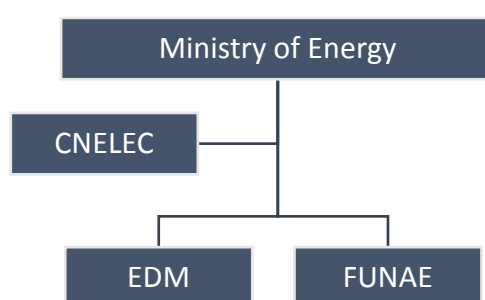
Figure 2. 19: Mozambique annual wind distribution. Source: (FUNAE, 2014)

The annual average of wind speed distribution for the entire country is estimated at 5 m/s. A larger caption for northern Mozambique shows indications of annual average wind speeds from 4 m/s to 6 m/s. (FUNAE, 2014)

It is believed that more precise and updated long-term measurements for the southern rim of the country should deliver wind speed readings of beyond 6 m/s on average, quite promising for commercial wind power deployment.

2.7.2 Institutional Framework of the Energy Sector

The institutional arrangement of the energy sector is organized as described in the following diagram:



The **Ministry of Energy** (MoE) is responsible for the coordination of all activities in the energy sector (regulatory framework), including electricity, biomass energy use, alternative technologies and downstream petroleum distribution and use.

Below of this ministry there are two main implementing institutions that carry out the activities of providing electricity to the population namely Mozambique Electricity company (EdM) and Energy Fund (FUNAE).

EdM is the national power utility, which has responsibility for the national grid (NG) network where it is involved in the electricity generation, transportation and distribution, collection of fees and in a small scale energy production through small hydro power plants and many off-grid generation in the case of remote villages that are not yet covered by the NG.

FUNAE is a Governmental Organization created in 1997 by ministerial decree. It is a public institution with administrative and financial autonomy, under the jurisdiction of the Minister of Energy. This organization is responsible for the dissemination and promotion of alternative energy technologies in rural areas that are not and will not be covered by the NG in the next ten years.

CNELEC, The National Electricity Council, is the institution that advises the Government and the MoE on tariff proposal and setting, as well as in the concession award process.

Actually the main energy systems that FUNAE provides for rural electrification are standalone PV systems, however, it is expected that in the near future a wide-scale installation of small local energy production devices of various types using different renewables resources will be promoted. Given that a study is being carried out by a hired company named GESTO to map out all available renewable resources in Mozambique, mainly solar and wind but also other local resources, some wider deployment efforts and larger investments in the field are pending.

To achieve this project, in 2011, several new measurement towers along the country were installed with solar and wind data measurement sensors, from which the data are being collected. The collected data will be crosschecked with available mesoscale and historical meteorological data from the National Institute of Meteorology (INAM) and the other institutions which are dealing with various natural resources. Some of the collected data will be used in the present work – namely the solar and wind resource data for the selected focus area in the southern edge of the country.

3 CHAPTER THREE: MODELLING SOFTWARE, LOCAL DATA AND RESEARCH METHODOLOGY

3.1 Specific Study Area Description

The selected study area/site is named “Estatuene Locality”, located in the south part of Mozambique, in Maputo Province, specifically in the Chagalane Administrative Post, Namaacha District. It is a small village very near to the border with Swaziland. It has 247 inhabitants, comprising a community of 52 rural households. The community of Estatuene is mainly smallholder farmers, where their main source of income is agriculture and livestock. The access to this village is by bulldozed dirt-roads, the residents do not have any access to electricity. Traditional energy sources such as candles, oil torches and biomass combustion/ fire wood are used for lighting and cooking (INE, 2007). Figure 3.1 shows the aerial satellite view of the village. The households are scattered and spread around an area with a main diameter of about 1000 meters.



Figure 3. 1: Satellite image of the village. Source: www.googleearth.com

As it can be seen the most concentrated houses are located within the circled area, and the households are living in a scattered manner being that only about 12 families are located quite nearby. In the present work we will focus only on this centrally located group of householders.

3.2 Modelling Software: HOMER

HOMER, *Hybrid Optimization Model for Electric Renewables* is a computer based program provided by the *National Renewable Energy Laboratory* (NREL) which can give assistance in the tasks to configure, simulate, evaluate and optimize several choices of designs of various systems for distributed electricity generation and consumption units.

The model inside the software helps with foreseeing the configuration of decentralized systems, and can combine conventional technologies and/or alternative ones: solar photovoltaic, wind, petrol or diesel

engines, biofuel, micro-turbines, small hydroelectric, thermoelectric, fuel cells, battery bank, among others evaluating a great number of alternatives in order to identify a system of lowest cost and capable to supply the demand of electricity to a certain community in an optimal way.

HOMER allows for a basic comparative analysis of several alternatives for electrification, and estimates impacts of changes in loading parameters, and environmental impact and the potential emissions of greenhouse gases.

Below are presented the potentialities of the HOMER software that are important for this study:

- Simulation of the system operation (energy balance for each of the 8760 hours of the year) comparing the energy demand and the capacity of the system to supply the energy on that hour, deciding for every hour how the generators operate and the loads of discharges of the batteries;
- Calculation of the energy balance for each proposed system that we want to consider, after the configuration is deemed feasible; and
- Estimating the costs of investment, equipment replacement, fuel, maintenance and operation.

The final goal is basically to identify the lowest cost system capable of meeting the electricity demand of a particular consumer unit, urban or rural residence, community, company or industry (HOMER, 2013).

3.3 Data and Modelling Approach

For the present work, the authors performed a literature research about the topic under study which is related to solar and wind energy sources, and devices involved in the conversion process of those resources into electricity for rural electrification of small villages through mini-grids using isolated solar-wind hybrid systems. It was also collected local site information from the respective authorities.

The technical data and characteristics of the equipment that would fit in the hybrid system, such as of wind turbines, photovoltaic panels, batteries bank for electricity/energy storage, converters/charge controllers, among others were obtained through technical catalogues of manufacturers, and internet surveys and contacts with some companies.

The HOMER Software helped in the specification, design and simulation of hybrid system for electricity generation to fully meet and comply with quality, continuity and security of the energy demand consumer electricity of the rural community in this study. For that reason, it was necessary to have the knowledge of the electricity demand of the study area, and the potential availability of solar radiation and wind from that area, which are described in the next section. The methodological approach applied was for electrification through a hybrid system combining photovoltaic and wind energy that consisted basically on the following steps:

- identification, field survey and analysis of energy demand of the rural community in study;

- analysis of the wind and solar radiation potential on that geographical region of the rural community using Excel sheets and HOMER software;
- analysis of the technical and financial feasibility of the hybrid system for electrification of the Estatuene rural area;
- with the aid of computer tools using HOMER Software was farther proceeded with simulations, comparative analysis of different configurations of hybrid systems and their costs in power generation to ensure the provision of electricity to the Estatuene rural community; and
- selection of the best configuration option of hybrid system, that meets quality and lowest cost and electricity demand of the Estatuene rural community.

3.3.1 Data Survey and Analysis of the Electric Power Curve of the Load Demand

The site location of the consumer was preceded by the completion of the mapping of the region through the collection of site geographical specification and checking the distance to the existing or expected electrical system. Taking into account that in the area there are 247 inhabitants or 52 rural households it was identified the consumption profile so that could be possible to draw the community daily curve load. This information can usually be found by measuring the consumption of electricity at the substation, but for the case of Estatuene village, there is no grid connection and no electricity power supply, thus the collection of that information were done through field surveys of possible power to be installed (lighting, sockets for general and specific use) and their periods of electricity use needs for each consumption unit, which summing all together would estimate the community consumption throughout the day, and then trace the curve of the representative daily load of the community.

According to the information received from the local authorities it was found that besides those 12 concentrated houses, the village comprises of public and private infrastructures constituted by one school, two churches, one bordering guard office and one trade center, which should be prioritized for the electrical supply. The final load data regarding the Estatuene energy demand was taken into account using the potential nearby infrastructures and their needs for electricity usage, which is mainly for illumination the above mentioned infrastructures, including the use of fridges, freezers for the conservation of vaccines and also for the use in communication devices such as radios, cell-phones and televisions (see Table 3.1). In this project the authors chose to add also a health center, a network of public illumination along the main street, and a public source of water (water pump) to be also electrified. The lifetime of the project is expected to be about 20 years.

The solar water pumping should be in DC given that it is cheaper and affordable than AC water pumping systems (PracticalAction, 2014).

Table 3. 1: Estatuene load description

Appliance	Unit Nominal Power (W)	1 School		2 Churches		1 security office		12 Households		1 Health Center		1 Trade Center		Public Water Well (Water Pump)		Public Illumination	
		Qty	Wp (Hr)	Qty	Wp (Hr)	Qty	Wp (Hr)	Qty	Wp (Hr)	Qty	Wp (Hr)	Qty	Wp (Hr)	Qty	Wp (Hr)	Qty	Wp (Hr)
Computer	250	2	07 AM-10 PM	---	---	1	07 AM-03 PM	4	07 AM -03 PM	---	---	---	---	---	---	---	---
Lamps	40	8	05 PM-11 PM	16	05 PM-11 PM	4	05 PM - 05 AM	36	05 PM-10 PM	8	05 PM-10 PM	3	05 PM-11PM	---	---	---	---
Fan	100	1	11 AM-04 PM	2	11 AM - 04 PM	1	11 AM -04 PM	6	11 AM-04 PM	1	11 AM-04 PM	1	11 AM-04 PM	---	---	---	---
TV	150	1	06 AM-11 PM	---	---	1	06 AM-11 PM	10	06 AM-11 PM	1	06 AM-11 PM	1	06 AM-11 PM	---	---	---	---
Radio	8	1	06 AM-11 PM	---	---	1	06 AM-11 PM	12	06 AM-11 PM	1	06 AM-11 PM	---	---	---	---	---	---
Freezer	400	---	---	---	---	---	---	3	(10/24)Hr	1	(10/24)Hr	1	(10/24)Hr	---	---	---	---
Fridge	200	---	---	---	---	1	(10/24)Hr	5	(10/24)Hr	1	(10/24)Hr	1	(10/24)Hr	---	---	---	---
HI-FI	70	---	---	2	08 AM-11 PM	---	---	3	08 AM -11 PM	---	---	1	08-23hr	---	---	---	---
Water Pump	720	---	---	---	---	---	---	---	---	---	---	---	---	1	06 AM -05 PM	---	---
Street Light	50	---	---	---	---	---	---	---	---	---	---	---	---	2	06 PM -05 AM	15	06 PM - 05 AM

As shown in table 3.1 above, the estimation of energy demand did not include appliances with electrical resistance heating such as irons, electric kettle, stove, etc., since their inclusion would lead to much higher power and system cost, therefore the community would be warned not to connect them in peak hours.

Table 3.2 and figure 3.2 below presents the daily load profile of the energy demand for the rural community of Estatuene village.

Table 3. 2: Estatuene Daily Energy Demand Profile

Hour	00-01	01-02	02-03	03-04	04-05	05-06	06-07	07-08	08-09	09-10	10-11	11-12
	1	2	3	4	5	6	7	8	9	10	11	12
Load - AC (kW)	2.51	2.51	2.51	2.51	2.51	3.72	3.72	5.47	5.89	5.89	5.89	7.09
Load - DC (kW)	-	-	-	-	0.72	0.72	0.72	0.72	0.72	0.72	0.72	0.72
Hour	12-13	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22-23	23-00
	13	14	15	16	17	18	19	20	21	22	23	24
Load - AC (kW)	7.09	7.09	7.09	5.84	4.64	8.49	8.49	8.49	8.49	8.49	6.23	2.51
Load - DC (kW)	0.72	0.72	0.72	0.72	-	-	-	-	-	-	-	-

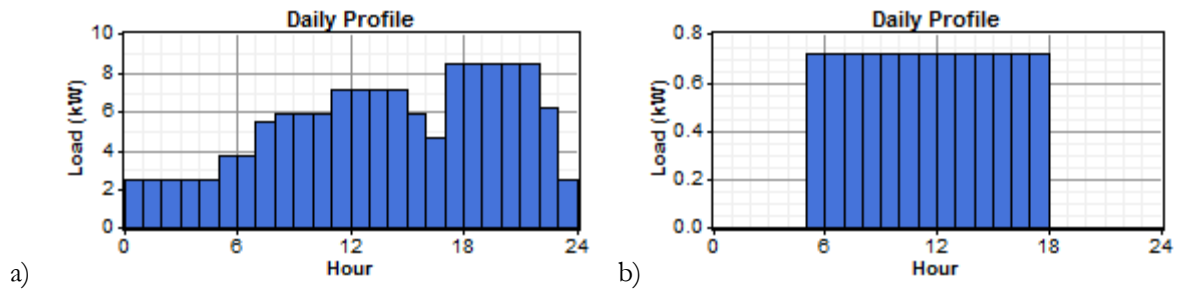


Figure 3. 2: Graphical view of daily energy demand: a) AC and b) DC Load, for the rural community of Estatuene.

The daily energy profile shows that DC charge demands 0.72 kW and remains constant from 04 AM to 04 PM hours, a period that is expected to have more water consumption in the locality. For the AC load, there is one consumption peak of around 8.49 kW, that is between 05 -10 PM hours, a period that almost all people are in their homes. The low consumption can be verified during the night where only street light and some appliances (fridge and freezers) are operating.

The experience gathered from FUNAE technicians based on the ongoing projects for rural electrification of 50 villages along the Mozambique countryside through standardized 4kW mini PV systems (see pictures of the system and typical electrified infrastructures/households in appendix 1), shows that most of rural inhabitants have low income, not being capable to acquire home appliances such as computers, fridges, refrigerators etc., therefore the estimates on energy demand have been determined taking that fact into account. Moreover, according to these technicians, some installed mini grids of 4kW are used to power about 15 households.

Therefore, the estimated average data of energy demand was introduced as input data in the HOMER software, for characterization of the electricity demand of the respective community in consideration.

3.3.2 Analysis of solar and wind potential at Estatuene Region

The wind data used in the study were collected from a meteorological station located 400 meters far from the village center which is operational since August 2012. Its geographical coordinates are: Latitude: 26°23'49.15"S and Longitude: 32° 5'30.45"E and 450 metres of altitude above mean sea level. Figure 3.3 shows the location map and the installed 60 meter tower/mast of the meteorological station.

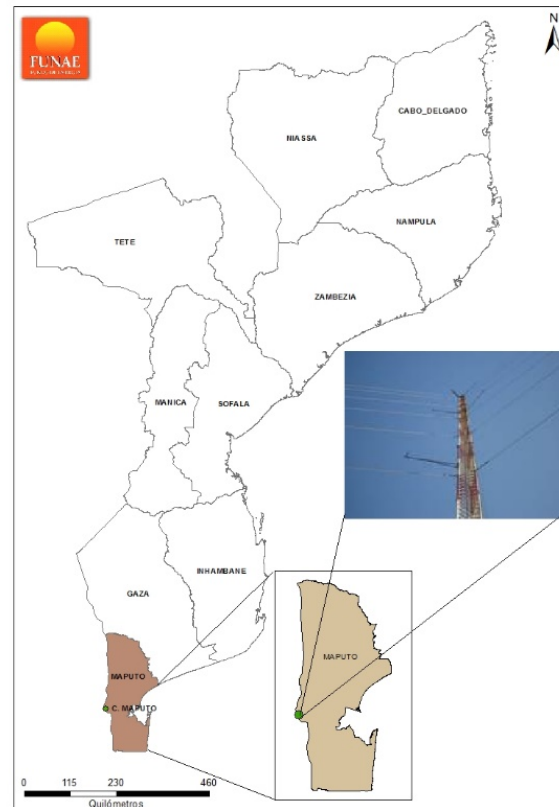


Figure 3. 3: Location map and measurement mast of the meteorological station

This meteorological station is composed of measurement sensors of wind speed and wind direction installed at 20, 40 and 60 meters respectively where the data are collected by averaging every 10 minutes and stored on its database. The station is programmed to automatically send data via global system for mobile (GSM) communication to FUNAE office collecting center once/twice a day.

The meteorological station does not hold a solar radiation sensor. Hence, the solar radiation data used in this study were obtained via internet databases, by introducing into HOMER the aforementioned site geographical coordinates. The solar and wind data of the region of Estatuene were collected and placed on HOMER platform as input data for resource specification and calculation. Computational simulations for the financial viability of possible hybrid configurations were therefore possible to be made.

a) Wind Data

For the present study, wind data collected at 20 m height was used, which were introduced in HOMER and with this software was possible to visualize the monthly wind speed distribution along the year, as presented in Figure 3.4 below.

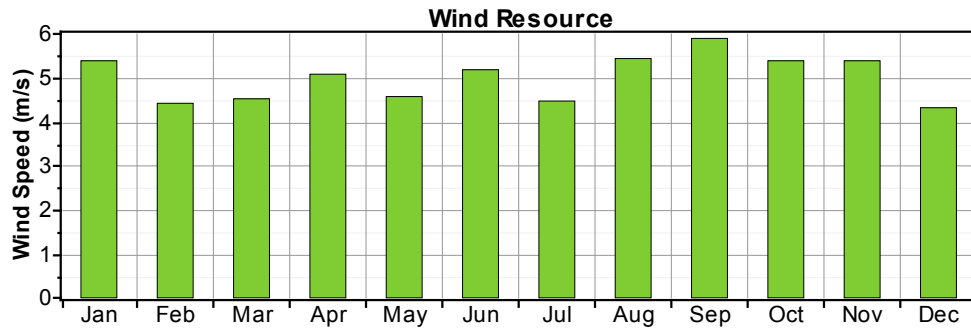


Figure 3. 4: Representation of the monthly mean wind speed (m/s) at 20 meters above ground level in the area of Estatuene.

According to the above figure in Estatuene region, the maximum estimated mean monthly wind speed is 5.92 m/s which can be observed in September, while a minimum value of 4.30 m/s can be observed in December; with an annual average of 5.02 m/s. Those wind speeds are feasible for wind energy production taking into account that most of the small-scale wind turbines have minimum start up wind speed around 2 m/s.

b) Solar Radiation Data

The monthly solar radiation given by HOMER software is depicted in figure 3.5 below.

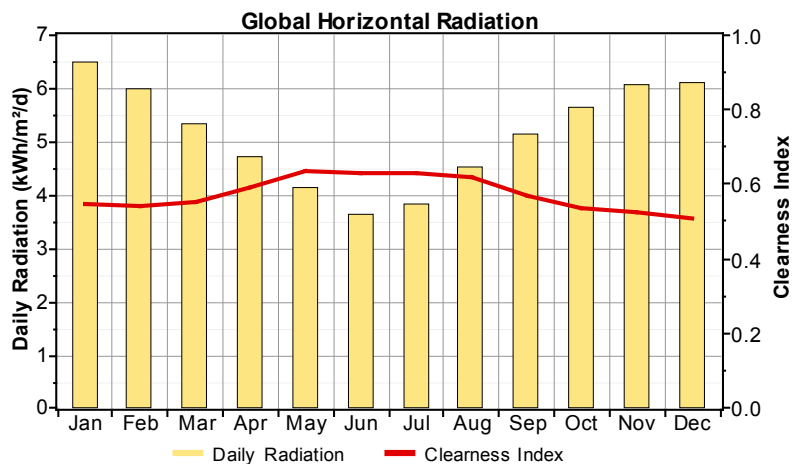


Figure 3. 5: Estimate of solar radiation potential of the area of Estatuene on 10/03/2014

The estimated solar radiation has a maximum value of 6.511 kWh/m²/day in January, the lowest value of 3.621 kWh/m²/day is in June, and an annual average of 5.13 kWh/m²/day. The software also introduces

the clearness⁴ index from the target site, showing high values during winter months (Apr-Aug) when the skies have clear conditions; and low values in the summer months (Sept-March) where most of the period are in cloudy conditions.

3.3.3 Analysis of Technological and Financial Resources used for the Feasibility Evaluation of Hybrid Systems

The present section describes the technological resources, their nominal power and costs used in the design through the simulations to supply electricity to the rural community of Estatuene.

a) Solar PV Modules.

For the proposed system there were chosen 150 Wp polycrystalline solar modules (specifications given in appendix 2) produced locally by the FUNAE factory. The cost per unit is around 193 USD. This figure was converted to cost per kWp in order to be introduced in HOMER software.

The operation and maintenance (O&M) costs were estimated as 1% of the total cost (Lenardic, 2014) which includes the equipment and installation costs (\$0.55/W) of the total cost (NREL, 2014).

The mounting slope angle for the solar panels is 26° oriented to the north (Azimuth 180°) as referred in the sections 2.1.7 conjugated with 3.3. The total capacity of the solar modules is obtained by HOMER simulations.

b) Wind Turbines.

Given that the AC and DC load demand of the whole system are approximately 8.4 kW and 0.7 kW respectively, to complement the solar PV system operation, there were chosen two wind turbines of 10kW and 5kW rated capacities (specifications given in appendix 3). However during the simulations HOMER will consider the turbine that best fits the system. The 10kW wind turbine (ref. WT FD8.0-10000) is supplied by Hikochi Electronics Limited – China, with CIF price of 21,052.62 USD and the 5kW wind turbine (ref. FD6.4-5000) is manufactured by Foshan Ouyad Electronic Co. Ltd, with CIF price of 10,526.32 USD (aliexpress, 2014). To those prices were added other costs such as transportation, due duties and installation that is 30% of the initial cost (AWEA, 2014).

Replacement costs was not taken into account as the lifetime of the wind turbine and solar modules are approximately 25 years and the system was basically designed for 20 years.

The wind turbine hub height is 12 meters as recommended by the manufacturer and these turbines rotate at 185° for FD8 and 200° for FD6 around their vertical axis facing the southeast direction from where it is the best wind flow, as referred in the wind rose diagram of figure 3.6.

⁴ Clearness index is the fraction of the solar radiation that is effectively transmitted through the atmosphere to strike the surface of the Earth. It varies between 0 and 1.

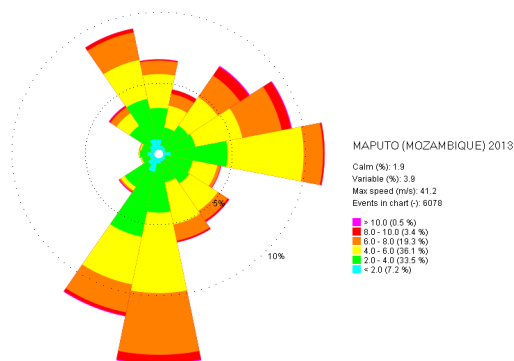


Figure 3. 6: Estatueene wind prevailing direction. Source: (Enviroware, 2014)

c) Storage System

The storage system for the present project is composed of batteries (OPZS3500 2V 3000Ah) (appendix 4), supplied by Hikochi Electronics Limited – China with CIF price of 1,200.00 USD per unit (aliexpress, 2014). The required number of batteries to meet the storage demand was given by HOMER simulations.

d) Converter

Various sizes of converters rated 5kW (ref. OF 5000) (appendix 5) as supplied by Hikochi Electronics Limited – China with CIF price of 2,432.57 USD per unit (aliexpress, 2014) were considered in the HOMER software for the designed system.

For both batteries and converters, the installation and O&M costs were calculated as it is done for solar modules. The replacement cost, which is the same as the initial cost, was considered taking into account the lifespan of about 10 years.

All the costs and technical information of the system components introduced in HOMER are summarized in the table 3.3 below.

Table 3. 3: Summary of technical and financial characteristics of technological resources used in the simulations

No	Equipment	Supplier	Total Cost/Unit	Total Cost/KW	Repl. Cost	O&M/Year
1	Batteries OPZS3500 2V 3000Ah	Hikochi Electronics Limited - China	1,576.00	n/a	1,576.00	15.76
2	FUNAE Solar Module, 160W	Mozambican Solar Panels - FUNAE	n/a	1,288.67	n/a	12.89
3	WT FD6.4-5000 --5KW	OUYAD	16,126.85	n/a	n/a	161.27
4	WT FD8.0-10000 --10KW	Hikochi Electronics Limited - China	32,253.67	n/a	n/a	322.54
5	Inverter/Controller SI 5000 ---- 5000W	Hikochi Electronics Limited - China	n/a	586.02	586.02	5.86

3.4 Specification of the configurations for the sensitivity analysis and simulations

After estimating the components of the above hybrid system, different sensitivity elements were introduced for the definition of the appropriate system for electrification of the site under study, which include:

- Minimum (3.6 m/s), mean (5.13 m/s) and maximum (6.5 m/s) values of the annual average wind speed;
- Minimum (4.4 kW/m²/d), mean (5.02 kW/m²/d) and maximum (5.98 kW/m²/d) values of the annual average solar radiation;
- Wind turbines with different capacities, as well as their supplied current type (DC or AC), quantities (0, 1, 2, 3, ..., 6) and different hub heights (12, 15 and 20 metres);
- Different capacity variation for the solar PV panels (0, 60, 65, ..., 68 kW) and their geographical orientation (26°);
- Different storage (batteries of 3000Ah) capacities (0, 1, 2, ..., 6 strings of 24 batteries); and
- Different capacities of converters (0, 5, 10, ..., 30kW) to feed the defined load.

Thus, through the computational tool “HOMER”, simulations of the configuration and combination of various options with their components of the hybrid system were made. After running the simulations, HOMER considered several sensitivity scenarios taking into account the involved system cost for electricity generation, the results of this being presented in the next chapter (Results and Discussion).

The results obtained in the sensitivity analysis were the basis for the selection of the best option scenario, prioritizing the costs of initial capital, operation and maintenance and the generated energy, that fit adequately to supply electricity for the rural community of Estatuene from the local load demand and the potential availability of solar and wind resources.

4 CHAPTER FOUR: RESULTS AND DISCUSSION

4.1 Optimization and Modelling

As stated earlier in chapter 3 the optimization model design process that was applied for the evaluation of the optimum choice for dimensioning of the hybrid system was the *Hybrid Optimization Model for Electric Renewables* (HOMER) version 2.81. The obtained results are presented and discussed in the sections that follow herein.

4.1.1 Hybrid Energy System Configuration

Figure 4.1 presents the overall component configuration of the designed system and its model in HOMER. The program set-up includes all the simulations and possible arrangements that were tested for solar PVs and wind turbines, for several sensitivity value ranges of generation capacity, financing costs, wind speed and solar irradiation as described in section 3.4 above.

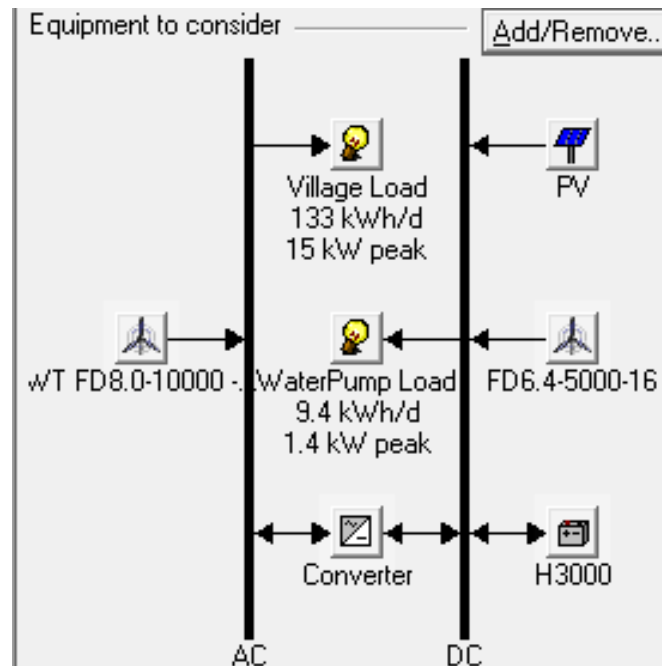


Figure 4. 1: Configuration of the system in HOMER

4.1.2 The Seasonal Load Demand Profile

As proposed, the village load demand profile is categorized in DC charges for water pumping, with a scaled annual average of 9.4 kWh/d and a peak load of 1.4kW; and AC charges for the remaining load with total scaled annual average of 133 kWh/d and a peak of 15.3 kW, which are presented in Figures 4.2.a and 4.2.b.

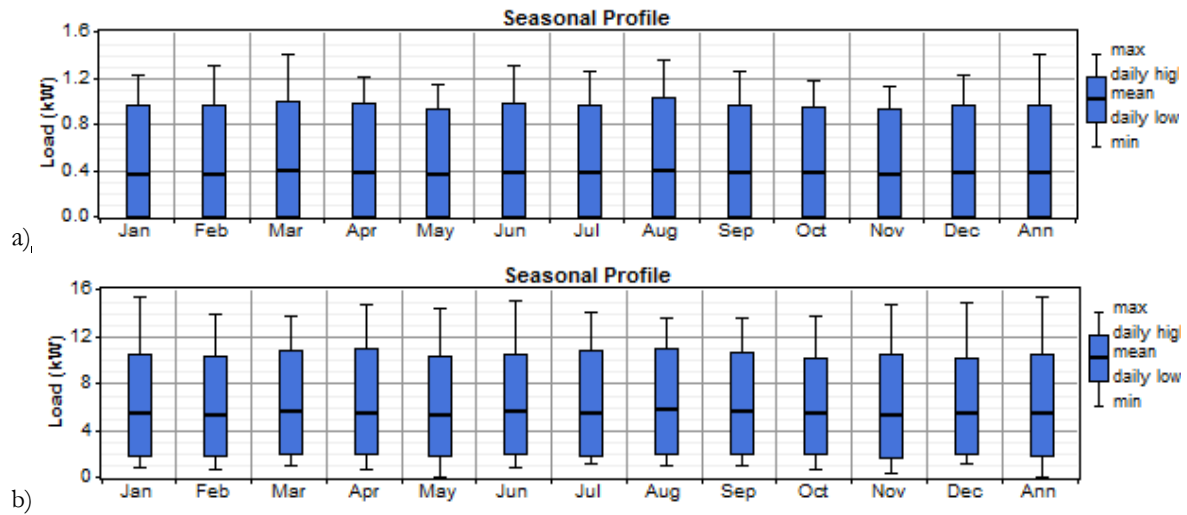


Figure 4. 2: Peak load profile for Estatuene Locality, a) DC load; b) AC load.

4.1.3 Results Obtained in the Simulations

Table 4.1 below summarizes the categorized configurations of the modelled system depicting **three main scenario variations (defined as SC1, SC2 and SC3)** for electricity generation. Please note that HOMER simulated the system for all proposed sensitivity variables as given in section 3.4, and the overall results are summarized in their entirety in Appendix 6. However, at the end the values that best fit the system's proposed parameters - being the mean values for solar radiation (5.13 kWh/m²/d), wind speed (5.02 m/s) and the minimum value of 12 meters for the hub height - were chosen as the representative ones for the main assessment criteria.

Table 4. 1: HOMER Simulation Categorized Results

Sensitivity Results

Optimization Results

Sensitivity variables

Global Solar (kWh/m²/d)

5.13

Wind Speed (m/s)

5.02

G1 Hub Height (m)

12

Double click on a system below for simulation results.

					PV (kW)	FD	G1	H3000	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Batt. Lf. (yr)
					66.0	1		48	15	\$ 185,783	3,198	\$ 226,661	0.341	1.00	20.0
					60.0			72	15	\$ 199,818	3,744	\$ 247,679	0.373	1.00	20.0
						8		192	15	\$ 440,974	8,178	\$ 545,520	0.821	1.00	20.0

Thus, the above scenarios resulted from the simulations are described in details below:

- SC1:** Hybrid (Solar-Wind) system composed of 66 kW PV panels, a 5 kW wind turbine, 48 batteries of 3,000Ah each and 3 inverters of 5kW each.
- SC2:** PV system alone composed of 60 kW PV panels, 72 batteries of 3,000Ah each and 3 inverters of 5kW each; and

- c) **SC3:** Wind system alone composed of 8 wind turbines of 5kW each, 192 batteries of 3,000Ah, and 3 inverters of 5kW each.

Table 4.2 below depicts the share of electricity production (in %) from each major system component for each of the main suggested scenarios.

Table 4. 2: Electrical energy production per each main scenario

	SC1		SC2		SC3	
Production	kWh	%	kWh	%	kWh	%
PV array	107,193.00	91	97,448.00	100	0	0
Wind turbine	10,231.00	9	0	0	81,847.00	100
Total	117,424.00	100	97,448.00	100	81,847.00	100

For each case scenario mentioned above, the following data were obtained from the simulation procedures and analysed further in the discussion:

- *Total net present cost (NPC⁵);*
- *Levelized cost of electricity (LCOE⁶); and*
- *Operating cost of electricity/ expenditures (OPE⁷).*

4.2 Techno-economic Analysis of the Results

In the techno-economic analysis of the project, each categorized scenario will be discussed in details regarding the technical and economy parameters resulting from the simulation procedures, and thereafter the best scenario that meets the demand at lowest costs will be chosen for further consideration.

4.2.1 Economic Analysis

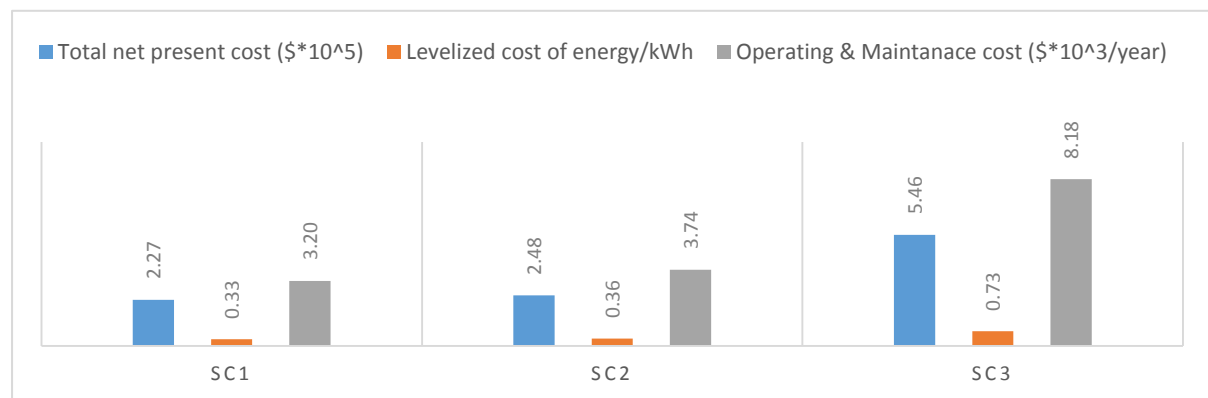


Figure 4. 3: Cost summary for the proposed system scenarios.

⁵ NPC is the present value of all the costs of installing and operating the system over the project lifetime, minus the present value of all the revenues that it earns over the project lifetime.

⁶ LCOE is the average cost per kWh of useful electrical energy produced by the system if it should be paying back for itself in reasonable economy terms.

⁷ OPE is the annualized value of all costs and revenues other than initial capital costs.

According to figure 4.3 above, we can see that the hybrid system scenario, SC1, presents low costs of O&M, LCOE and NPC compared to the other scenarios of separated wind or solar systems alone. Detailed analysis of the cash flow summary (figure 4.4) shows that the costs of the batteries are relatively high compared to other devices in the system.

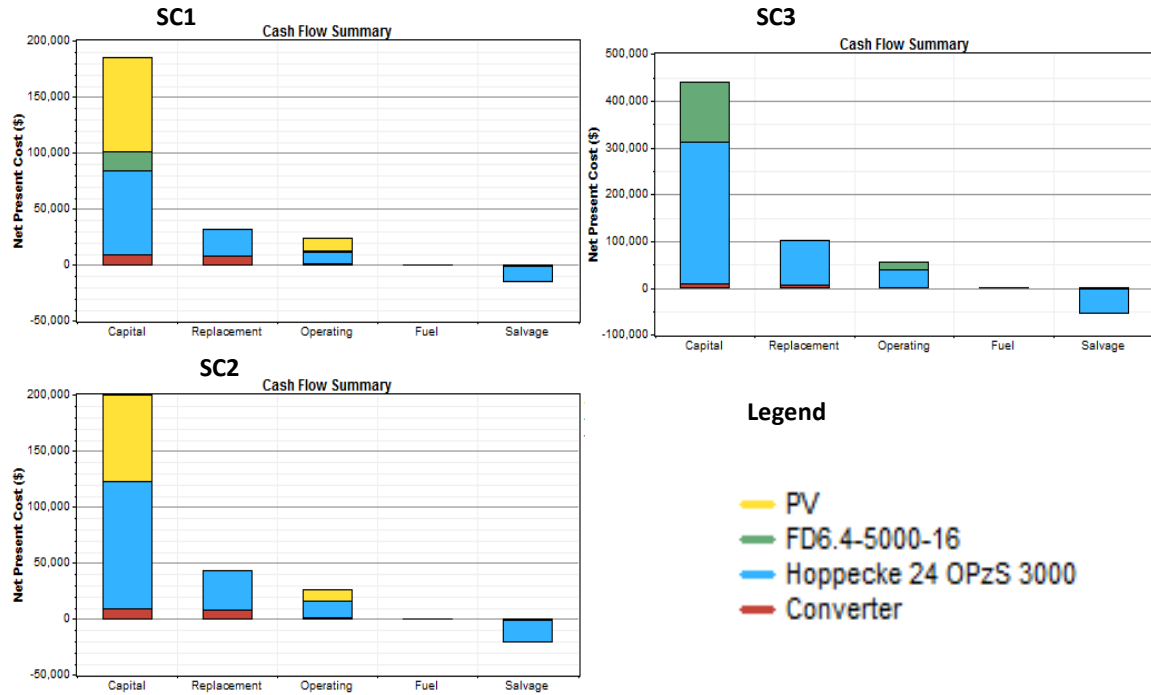


Figure 4. 4: Cash flow summary for the three selected scenarios.

The capital, replacement and operating costs of batteries from SC1 tends to be lower compared to both SC2 and SC3 however, the SC1 salvage⁸ is lower than other scenarios given its low storage capacity. Salvage is the remaining value in a component of the power system at the end of the project lifetime. More details are given in the technical analysis section 4.2.2.

4.2.2 Technical Analysis

Additional analysis needs to be performed for the system feasibility and technical performance, such as the annual electricity production⁹, consumption¹⁰ and autonomy¹¹ parameters of each scenario as shown in figure 4.5 below. System autonomy is the ratio of the battery bank capacity to the electric load size.

⁸ Salvage is the remaining value in a component of the power system at the end of the project lifetime.

⁹ The total electrical production is the total amount of electrical energy produced by the power system in one year. It is the sum of the electrical energy produced by all components of the system.

¹⁰ The electricity consumption is the total amount of electrical energy that went to serve each of the system's electrical loads.

¹¹ The autonomy is the ratio of the battery bank size to the electric load.

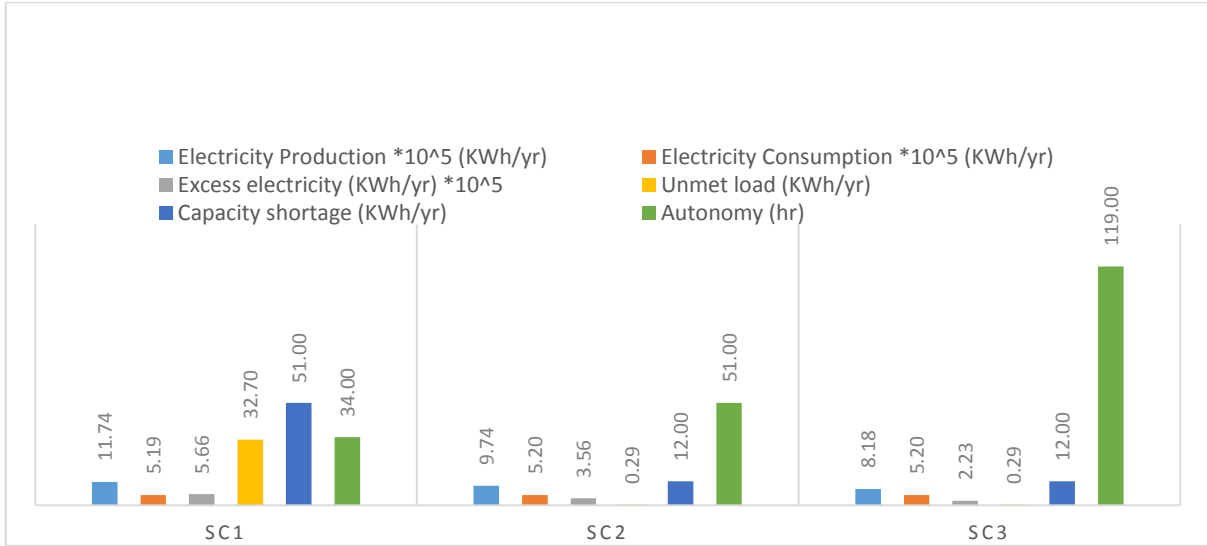


Figure 4. 5: Technical evaluation results

According to the overall results presented in figure 4.5, the hybrid scenario SC1 produces most electricity compared to both SC2 and SC3 due to the combination of two sources of energy. That fact leads to complementarity of resources in cases of absence of one or low availability of either the resource potential or the technical readiness of the system components.

However, SC1 presents a relatively low consumption compared to both SC2 and SC3, given that during a greater period of the year, the energy produced by this system cannot be stored due to low capacity of the storage bank (fewer batteries) and this fact leads to higher **unmet load**¹² demand and a considerable **capacity shortage**¹³ of the hybrid scenario.

Moreover, SC1 has a larger amount of **excess of electricity**¹⁴ that is lost in relation to other scenarios due to its lesser storage system and leading to a relatively low level of **autonomy** of the system.

In off-grid energy systems the most sensitive component are the batteries (energy storage), as they are relatively more expensive and need special attention regarding maintenance and monitoring of quality of charge/discharge cycling.

Figure 4.6 below shows an overview of the battery performance and state of charge during one year for each scenario. It can be observed that the batteries remain most of the working period charged at the recommended levels (above 40% state of charge or 60% DoD) as mentioned in 2.3.2 d). However, in SC1 several critical points are observed during the months of March, June and August where the batteries reach a state of charge below 40% due to low solar radiation during these months, even though wind speeds are comparatively high. The other two scenarios in the analysis show some single and less challenging critical points for the batteries in August and April, respectively.

¹² The unmet load is the proportion of the total load that went unserved because of insufficient generation.

¹³ The capacity shortage is a shortfall that occurs between the required operating capacity and the actual amount of operating capacity the system can provide.

¹⁴ Excess electricity is surplus electrical energy that must be dumped because it cannot be used to serve a load or charge the batteries.

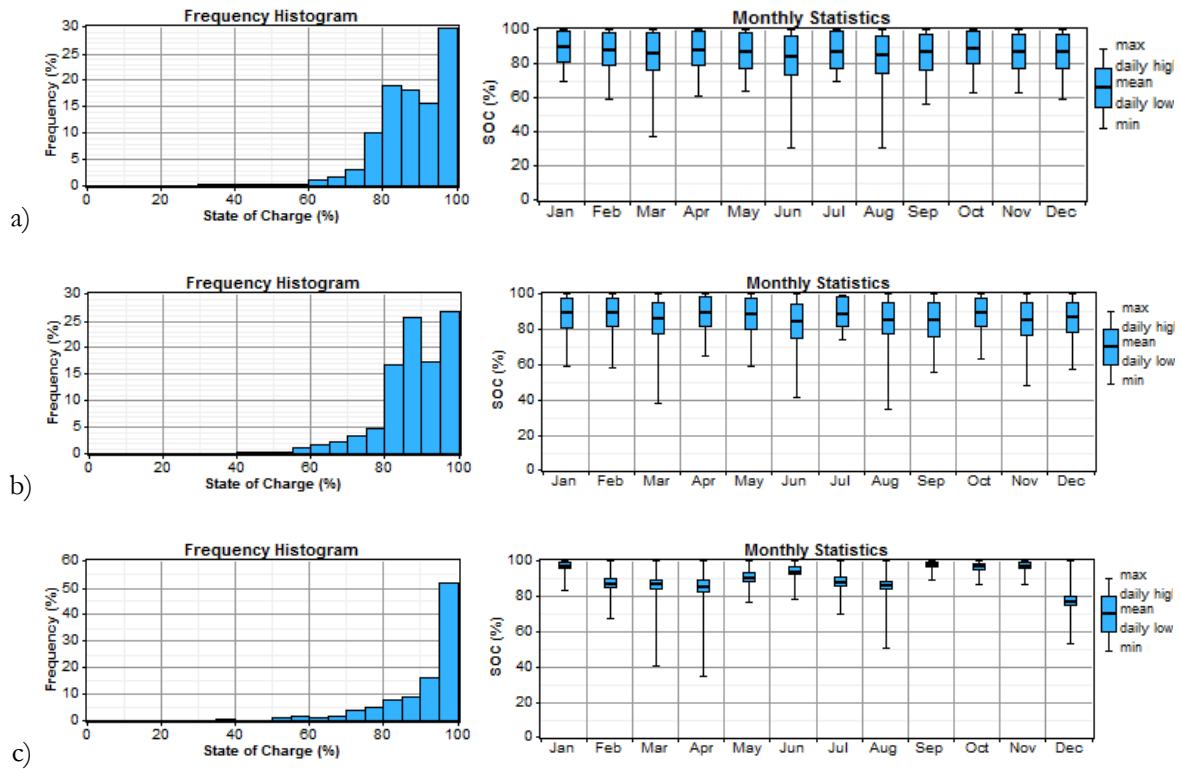


Figure 4. 6: Battery state of charge a) SC1; b) SC2; c) SC3

To overcome this problem, the installation of a remote control system to monitor the batteries' state of charge has been suggested so that in case of dangerous DoD levels, the beneficiaries can be warned to reduce electricity usage by disconnecting some appliances that are not relevant in a specific point of time or totally switch off the electricity supply to the householders and leave only important infrastructures connected, such as schools, public lights, health centre, water pump etc, until the batteries are recharged.

Through the techno-economic analysis, it can be noted that although the SC1 have high values of excess electricity leading to electricity losses, most of the time the system is incapable to satisfy the load demand (unmet demand) and experiences more critical points of DoD compared to the other two scenarios, the SC1 remains the best option as it can meet the load at a relatively lowest cost.

Furthermore, the excess electricity produced by the SC1 system can be used in the future to connect more beneficiaries simply by adding more batteries and expanding the system boundaries, given that with the electrification of the village conditions will be created for enhanced local development and it is predicted that scattered people/householders will come closer seeking for electricity and availing of the related development possibilities.

If the demand continues increasing, the SC1 system remains feasible given the modularity of equipment structure. Or in other words, more batteries or more PV panels, wind turbines, etc, can be gradually added to increase electricity production successively.

5 CHAPTER FIVE: CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

The present work proposes the need for and performs a detailed feasibility study for implementing wind and solar hybrid power system for rural electrification at the Estatuene Locality in the southern part of Mozambique. The results of the assessment made on the availability of energy resources in the study area show promising data for the implementation of the proposed system, as the annual average potential is approximately 5.02 m/s for the wind velocity resource and 5.13 kWh/m²/day for the solar insolation resource, respectively.

The feasibility analysis was attempted in order to find the most appropriate system for the electrification of Estatuene combining the two energy sources (solar and wind) using the HOMER software. All system scenarios need also to contain a large battery bank to cover the necessary demand.

The pre-evaluation considerations resulted in three optimal options/scenarios to be compared among each-other in terms of performance and costs, namely:

- Solar-wind hybrid energy system,
- Solar PV energy system alone, and
- Wind turbine energy system alone.

However, once a critical analysis was made between these options it was concluded that the hybrid combination appears to be more technically and economically feasible, because the costs of electricity production using this system are lowest (0.34 \$/kWh) compared to the other two options (0.37 \$/kWh for solar-only and 0.82 \$/kWh for wind-only) and also involves less equipment especially in terms of energy storage devices (battery charge).

Despite the fact that the levelized cost of electricity for the hybrid system is lower, it is still an unbearably high investment for the economic capacity of rural communities in Mozambique, since even in urban areas that are covered by the national grid (supplied by EDM), the cost of electricity for domestic use is about 0.1 \$USD/kWh. However, it is the government responsibility to supply electricity to citizens. So for this kind of projects, the Government of Mozambique and cooperating partners provide funds for implementation of rural electrification projects through FUNAE, and then it is up to the involved communities to contribute about 15 USD/Month/beneficiary and ensure the necessary local maintenance and operation-supporting procedures to keep the system properly running and to sustain the longevity of the equipment.

In Mozambique the measures for rural electrification funded by the Government through FUNAE are the implementation of small renewable energy systems, especially using PVs, but this technology has been

costly and technically impractical because these systems, in certain periods of the day or month, do not support the load demand due to insufficient availability of the resource, in turn requiring a prohibitively large energy storage auxiliary systems which further raise the costs. This situation could be overcome through the implementation of hybrid (wind-solar) power systems.

5.2 Recommendations and future work

It is recommended that:

- Studies of this nature be transferred and implemented also in other parts of the country, to determine the feasibility for implementation of hybrid projects using renewables elsewhere in Mozambique; and
- The Estatuene project should be implemented in a pilot phase, so that later on it can be replicated to other regions of the country that yet have not been covered by the national grid authority EDM;
- Future socio-economic analysis must be made to evaluate the possibility so that the rural communities should pay the regular EDM electricity tariff (0.1 USD/kWh) and the remaining production cost (LCOE) inferred by renewable energy generation should be subsidized by the Government.

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APPENDICES

Appendix 1: FUNAE 4kW mini PV systems & typical electrified infrastructures/households.



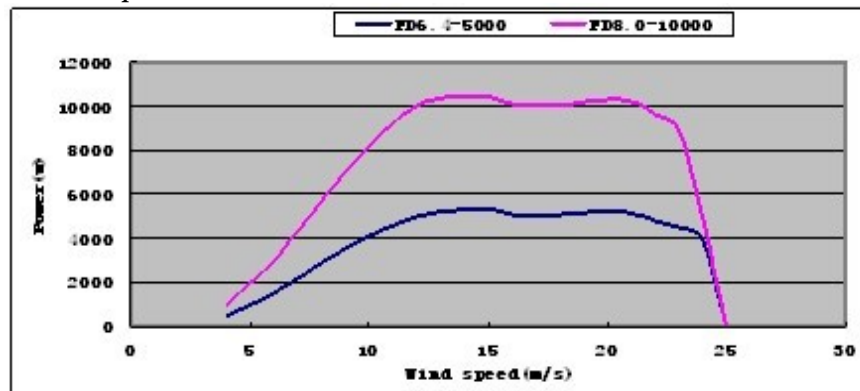
Appendix 2: FUNAE Solar modules Characteristics

Technical Specifications	
Power (W)	150
V _{mpp} (V)	17.14
I _{mpp} (A)	8.17
V _{ca} (V)	22.14
I _{sc} (A)	8.69
Tolerance (%)	2.5
V _{max} (V)	1000.0
Efficiency (%)	16.60

Appendix 3: Wind Turbines - FD6.4-5000 and FD8.0-10000 Characteristics

a) Technical Specifications		
Parameter:	FD6.4-5000 (DC)	FD8.0-10000 (AC)
Rated Power (W)	5000	10000
Rated Voltage (V)	240	240
Rotor diameter (m)	5.4	8
Start-up wind speed (m/s)	2	2.5
Rated wind speed (m/s)	12	16
Safety wind speed (m/s)	60	60
Rotating rate (r/m)	200	45
Blade material	Fiber glass	Reinforced fiber glass
Blade quantity	3	3
Tower height (m)	12	12
Tower diameter (mm)	273	325
Matched inverter type	Sine wave	Sine wave (can choose 3 phase)

b) Wind Turbines' power curves:



Appendix 4: 5000W off-grid, pure sine wave inverter, for solar and wind system.



Technical Specifications	
Model Number	OF5000
Type	Pure sine wave off grid inverter
Output Power (W)	5000
Input Voltage (v)	48/96
Output Current (A)	15
Output Voltage (V)	220/110
Output Frequency (Hz)	50/60

Appendix 5: Batteries (2v, OPZS3500, 3000Ah)



Capacities, discharging and charging rates for OPzS batteries with hard rubber container

Discharge time (h)	Capacity at 20° in Ah				Discharge rate in A				Charging current in A		
	10	5	3	1	10	5	3	1	to Gas-evolution	from Gas-evolution	falling to
Final voltage V/ C	1.80	1.77	1.75	1.67	1.80	1.77	1.75	1.67		falling from	
Celltype											
14 OPzS 3500	3500.0	2940.0	2520.0	1680.0	350.0	588.0	840.0	1680.0	525.0	232.0	116.0

Appendix 6: Overall optimization results in HOMER

PV (kW)	FD 6.4 5000	FD8. 10000	H3000	Converter (kW)	Initial capital	Operating cost (\$/yr)	Total NPC	COE (\$/kWh)	Renewable fraction	Batt. Lf. (yr)
66	1		48	15	\$185,783	3,198	\$226,661	0.341	1	20
66.5	1		48	15	\$186,428	3,204	\$227,389	0.343	1	20
67	1		48	15	\$187,072	3,211	\$228,116	0.344	1	20
67.5	1		48	15	\$187,717	3,217	\$228,844	0.345	1	20
68	1		48	15	\$188,361	3,224	\$229,572	0.346	1	20
65	1		48	20	\$187,424	3,388	\$230,728	0.348	1	20
65.5	1		48	20	\$188,069	3,394	\$231,456	0.349	1	20
66	1		48	20	\$188,713	3,401	\$232,183	0.35	1	20
66.5	1		48	20	\$189,358	3,407	\$232,911	0.351	1	20
67	1		48	20	\$190,002	3,414	\$233,638	0.352	1	20
67.5	1		48	20	\$190,647	3,420	\$234,366	0.353	1	20
68	1		48	20	\$191,291	3,427	\$235,093	0.354	1	20
60	2		48	15	\$194,176	3,281	\$236,115	0.356	1	20
65	1		48	25	\$190,354	3,590	\$236,250	0.356	1	20
65.5	1		48	25	\$190,999	3,597	\$236,977	0.357	1	20
66	1		48	25	\$191,643	3,603	\$237,705	0.358	1	20
66.5	1		48	25	\$192,288	3,610	\$238,433	0.359	1	20
67	1		48	25	\$192,932	3,616	\$239,160	0.36	1	20
67.5	1		48	25	\$193,577	3,623	\$239,888	0.361	1	20
68	1		48	25	\$194,221	3,629	\$240,615	0.362	1	20
60	2		48	20	\$197,106	3,484	\$241,637	0.364	1	20
65	1		48	30	\$193,284	3,793	\$241,772	0.364	1	20
65.5	1		48	30	\$193,929	3,800	\$242,499	0.365	1	20
66	1		48	30	\$194,573	3,806	\$243,227	0.366	1	20
65	2		48	15	\$200,621	3,346	\$243,391	0.366	1	20
66.5	1		48	30	\$195,218	3,813	\$243,954	0.367	1	20
65.5	2		48	15	\$201,266	3,352	\$244,119	0.368	1	20
67	1		48	30	\$195,862	3,819	\$244,682	0.369	1	20
66	2		48	15	\$201,910	3,359	\$244,846	0.369	1	20
67.5	1		48	30	\$196,507	3,826	\$245,410	0.37	1	20
66.5	2		48	15	\$202,555	3,365	\$245,574	0.37	1	20
68	1		48	30	\$197,151	3,832	\$246,137	0.371	1	20
67	2		48	15	\$203,199	3,372	\$246,302	0.371	1	20
67.5	2		48	15	\$203,844	3,378	\$247,029	0.372	1	20
60	2		48	25	\$200,036	3,686	\$247,159	0.372	1	20
60			72	15	\$199,818	3,744	\$247,679	0.373	1	20
68	2		48	15	\$204,488	3,385	\$247,757	0.373	1	20
65	2		48	20	\$203,551	3,549	\$248,913	0.375	1	20
65.5	2		48	20	\$204,196	3,555	\$249,641	0.376	1	20
66	2		48	20	\$204,840	3,562	\$250,368	0.377	1	20
66.5	2		48	20	\$205,485	3,568	\$251,096	0.378	1	20

67	2		48	20	\$206,129	3,575	\$251,823	0.379	1	20
67.5	2		48	20	\$206,774	3,581	\$252,551	0.38	1	20
60	2		48	30	\$202,966	3,889	\$252,681	0.381	1	20
60			72	20	\$202,748	3,947	\$253,201	0.381	1	20
68	2		48	20	\$207,418	3,588	\$253,279	0.381	1	20
60	3		48	15	\$210,303	3,442	\$254,300	0.383	1	20
65	2		48	25	\$206,481	3,751	\$254,435	0.383	1	20
65			72	15	\$206,263	3,809	\$254,955	0.384	1	20
65.5	2		48	25	\$207,126	3,758	\$255,162	0.384	1	20
65.5			72	15	\$206,908	3,816	\$255,683	0.385	1	20
66	2		48	25	\$207,770	3,764	\$255,890	0.385	1	20
66			72	15	\$207,552	3,822	\$256,410	0.386	1	20
66.5	2		48	25	\$208,415	3,771	\$256,618	0.386	1	20
66.5			72	15	\$208,197	3,829	\$257,138	0.387	1	20
67	2		48	25	\$209,059	3,777	\$257,345	0.387	1	20
67			72	15	\$208,841	3,835	\$257,866	0.388	1	20
67.5	2		48	25	\$209,704	3,784	\$258,073	0.389	1	20
67.5			72	15	\$209,486	3,842	\$258,593	0.389	1	20
60			72	25	\$205,678	4,150	\$258,723	0.39	1	20
68	2		48	25	\$210,348	3,790	\$258,800	0.39	1	20
68			72	15	\$210,130	3,848	\$259,321	0.39	1	20
60	3		48	20	\$213,233	3,645	\$259,822	0.391	1	20
65	2		48	30	\$209,411	3,954	\$259,957	0.391	1	20
65			72	20	\$209,193	4,012	\$260,477	0.392	1	20
65.5	2		48	30	\$210,056	3,961	\$260,684	0.392	1	20
65.5			72	20	\$209,838	4,018	\$261,205	0.393	1	20
66	2		48	30	\$210,700	3,967	\$261,412	0.394	1	20
65	3		48	15	\$216,748	3,507	\$261,576	0.394	1	20
66			72	20	\$210,482	4,025	\$261,932	0.394	1	20
66.5	2		48	30	\$211,345	3,974	\$262,140	0.395	1	20
65.5	3		48	15	\$217,393	3,513	\$262,304	0.395	1	20
66.5			72	20	\$211,127	4,031	\$262,660	0.395	1	20
67	2		48	30	\$211,989	3,980	\$262,867	0.396	1	20
66	3		48	15	\$218,037	3,520	\$263,032	0.396	1	20
67			72	20	\$211,771	4,038	\$263,387	0.397	1	20
67.5	2		48	30	\$212,634	3,987	\$263,595	0.397	1	20
66.5	3		48	15	\$218,682	3,526	\$263,759	0.397	1	20
67.5			72	20	\$212,416	4,044	\$264,115	0.398	1	20
60			72	30	\$208,608	4,352	\$264,245	0.398	1	20
68	2		48	30	\$213,278	3,993	\$264,322	0.398	1	20
67	3		48	15	\$219,326	3,533	\$264,487	0.398	1	20
68			72	20	\$213,060	4,051	\$264,843	0.399	1	20
67.5	3		48	15	\$219,971	3,539	\$265,214	0.399	1	20
60	3		48	25	\$216,163	3,847	\$265,344	0.399	1	20
60	1		72	15	\$215,945	3,905	\$265,864	0.4	1	20

68	3		48	15	\$220,615	3,546	\$265,942	0.4	1	20
65			72	25	\$212,123	4,215	\$265,999	0.4	1	20
65.5			72	25	\$212,768	4,221	\$266,726	0.402	1	20
65	3		48	20	\$219,678	3,710	\$267,098	0.402	1	20
66			72	25	\$213,412	4,228	\$267,454	0.403	1	20
65.5	3		48	20	\$220,323	3,716	\$267,826	0.403	1	20
66.5			72	25	\$214,057	4,234	\$268,182	0.404	1	20
66	3		48	20	\$220,967	3,723	\$268,553	0.404	1	20
67			72	25	\$214,701	4,241	\$268,909	0.405	1	20
66.5	3		48	20	\$221,612	3,729	\$269,281	0.405	1	20
67.5			72	25	\$215,346	4,247	\$269,637	0.406	1	20
67	3		48	20	\$222,256	3,736	\$270,009	0.406	1	20
68			72	25	\$215,990	4,254	\$270,364	0.407	1	20
67.5	3		48	20	\$222,901	3,742	\$270,736	0.408	1	20
60	3		48	30	\$219,093	4,050	\$270,866	0.408	1	20
60	1		72	20	\$218,875	4,108	\$271,386	0.409	1	20
68	3		48	20	\$223,545	3,749	\$271,464	0.409	1	20
65			72	30	\$215,053	4,417	\$271,521	0.409	1	20
65.5			72	30	\$215,698	4,424	\$272,248	0.41	1	20
60	4		48	15	\$226,430	3,603	\$272,486	0.41	1	20
65	3		48	25	\$222,608	3,912	\$272,620	0.41	1	20
66			72	30	\$216,342	4,430	\$272,976	0.411	1	20
65	1		72	15	\$222,390	3,970	\$273,140	0.411	1	20
65.5	3		48	25	\$223,253	3,919	\$273,348	0.412	1	20
66.5			72	30	\$216,987	4,437	\$273,704	0.412	1	20
65.5	1		72	15	\$223,035	3,977	\$273,868	0.412	1	20
66	3		48	25	\$223,897	3,925	\$274,075	0.413	1	20
67			72	30	\$217,631	4,443	\$274,431	0.413	1	20
66	1		72	15	\$223,679	3,983	\$274,596	0.413	1	20
66.5	3		48	25	\$224,542	3,932	\$274,803	0.414	1	20
67.5			72	30	\$218,276	4,450	\$275,159	0.414	1	20
66	6		24	20	\$231,452	3,420	\$275,174	0.415	1	20
66.5	1		72	15	\$224,324	3,990	\$275,323	0.414	1	20
67	3		48	25	\$225,186	3,938	\$275,530	0.415	1	20
68			72	30	\$218,920	4,456	\$275,886	0.415	1	20
66.5	6		24	20	\$232,097	3,427	\$275,902	0.416	1	20
67	1		72	15	\$224,968	3,996	\$276,051	0.416	1	20
67.5	3		48	25	\$225,831	3,945	\$276,258	0.416	1	20
67	6		24	20	\$232,741	3,433	\$276,630	0.417	1	20
67.5	1		72	15	\$225,613	4,003	\$276,778	0.417	1	20
60	1		72	25	\$221,805	4,311	\$276,908	0.417	1	20
68	3		48	25	\$226,475	3,951	\$276,986	0.417	1	20
67.5	6		24	20	\$233,386	3,440	\$277,357	0.418	1	20
68	1		72	15	\$226,257	4,009	\$277,506	0.418	1	20
60	4		48	20	\$229,360	3,806	\$278,007	0.419	1	20

68	6		24	20	\$234,030	3,446	\$278,085	0.419	1	20
65	3		48	30	\$225,538	4,115	\$278,142	0.419	1	20
65	1		72	20	\$225,320	4,173	\$278,662	0.42	1	20
65.5	3		48	30	\$226,183	4,122	\$278,869	0.42	1	20
60	7		24	15	\$236,915	3,301	\$279,107	0.42	1	20
65.5	1		72	20	\$225,965	4,179	\$279,390	0.421	1	20
66	3		48	30	\$226,827	4,128	\$279,597	0.421	1	20
65	4		48	15	\$232,875	3,668	\$279,761	0.421	1	20
65		2	48	15	\$232,875	3,670	\$279,787	0.422	1	20
66	1		72	20	\$226,609	4,186	\$280,117	0.422	1	20
66.5	3		48	30	\$227,472	4,135	\$280,325	0.422	1	20
65.5	4		48	15	\$233,520	3,674	\$280,489	0.422	1	20
65.5		2	48	15	\$233,520	3,676	\$280,515	0.423	1	20
66	6		24	25	\$234,382	3,623	\$280,696	0.423	1	20
66.5	1		72	20	\$227,254	4,192	\$280,845	0.423	1	20
67	3		48	30	\$228,116	4,141	\$281,052	0.423	1	20
66	4		48	15	\$234,164	3,681	\$281,217	0.423	1	20
66		2	48	15	\$234,164	3,683	\$281,242	0.424	1	20
66.5	6		24	25	\$235,027	3,630	\$281,424	0.424	1	20
67	1		72	20	\$227,898	4,199	\$281,573	0.424	1	20
67.5	3		48	30	\$228,761	4,148	\$281,780	0.424	1	20
66.5	4		48	15	\$234,809	3,687	\$281,944	0.424	1	20
66.5		2	48	15	\$234,809	3,689	\$281,970	0.425	1	20
67	6		24	25	\$235,671	3,636	\$282,151	0.425	1	20
67.5	1		72	20	\$228,543	4,205	\$282,300	0.425	1	20
60	1		72	30	\$224,735	4,513	\$282,430	0.425	1	20
68	3		48	30	\$229,405	4,154	\$282,507	0.425	1	20
67	4		48	15	\$235,453	3,694	\$282,672	0.426	1	20
67		2	48	15	\$235,453	3,696	\$282,697	0.426	1	20
67.5	6		24	25	\$236,316	3,643	\$282,879	0.426	1	20
68	1		72	20	\$229,187	4,212	\$283,028	0.426	1	20
67.5	4		48	15	\$236,098	3,700	\$283,399	0.427	1	20
67.5		2	48	15	\$236,098	3,702	\$283,425	0.427	1	20
60	4		48	25	\$232,290	4,008	\$283,529	0.427	1	20
68	6		24	25	\$236,960	3,649	\$283,607	0.427	1	20
60	2		72	15	\$232,072	4,066	\$284,050	0.428	1	20
60		1	72	15	\$232,072	4,067	\$284,062	0.428	1	20
68	4		48	15	\$236,742	3,707	\$284,127	0.428	1	20
68		2	48	15	\$236,742	3,709	\$284,153	0.428	1	20
65	1		72	25	\$228,250	4,376	\$284,184	0.428	1	20
60	7		24	20	\$239,845	3,503	\$284,628	0.429	1	20
65.5	1		72	25	\$228,895	4,382	\$284,912	0.429	1	20
65	4		48	20	\$235,805	3,871	\$285,283	0.429	1	20
65		2	48	20	\$235,805	3,873	\$285,309	0.43	1	20
66	1		72	25	\$229,539	4,389	\$285,639	0.43	1	20

65.5	4		48	20	\$236,450	3,877	\$286,011	0.431	1	20
65.5		2	48	20	\$236,450	3,879	\$286,036	0.431	1	20
66	6		24	30	\$237,312	3,826	\$286,218	0.431	1	20
66.5	1		72	25	\$230,184	4,395	\$286,367	0.431	1	20
65	7		24	15	\$243,360	3,366	\$286,383	0.431	1	20
66	4		48	20	\$237,094	3,884	\$286,738	0.432	1	20
66		2	48	20	\$237,094	3,886	\$286,764	0.432	1	20
66.5	6		24	30	\$237,957	3,832	\$286,946	0.432	1	20
67	1		72	25	\$230,828	4,402	\$287,094	0.432	1	20
65.5	7		24	15	\$244,005	3,372	\$287,110	0.432	1	20
66.5	4		48	20	\$237,739	3,890	\$287,466	0.433	1	20
66.5		2	48	20	\$237,739	3,892	\$287,492	0.433	1	20
67	6		24	30	\$238,601	3,839	\$287,673	0.433	1	20
67.5	1		72	25	\$231,473	4,408	\$287,822	0.433	1	20
66	7		24	15	\$244,649	3,379	\$287,838	0.433	1	20
67	4		48	20	\$238,383	3,897	\$288,194	0.434	1	20
67		2	48	20	\$238,383	3,899	\$288,219	0.434	1	20
67.5	6		24	30	\$239,246	3,845	\$288,401	0.434	1	20
68	1		72	25	\$232,117	4,415	\$288,550	0.434	1	20
66.5	7		24	15	\$245,294	3,385	\$288,565	0.435	1	20
67.5	4		48	20	\$239,028	3,903	\$288,921	0.435	1	20
67.5		2	48	20	\$239,028	3,905	\$288,947	0.435	1	20
60	4		48	30	\$235,220	4,211	\$289,051	0.435	1	20
68	6		24	30	\$239,890	3,852	\$289,129	0.436	1	20

Appendix 7: Scenario 1, 2 and 3 System Report

SC1 – System Report

Sensitivity case

Solar Data Scaled Average: 5.13 kWh/m²/d

Wind Data Scaled Average: 5.02 m/s

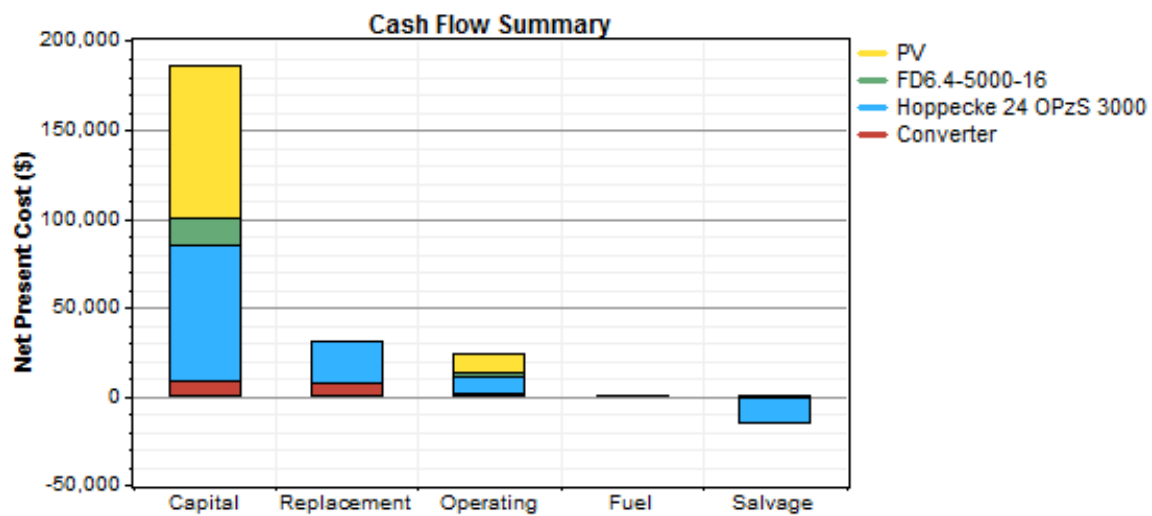
WT FD8.0-10000 --10KW Hub Height: 12 m

System architecture

PV Array	66 kW
Wind turbine 1	FD6.4-5000-16
Battery	48 Hoppecke 24 OPzS 3000
Inverter	15 kW
Rectifier	15 kW

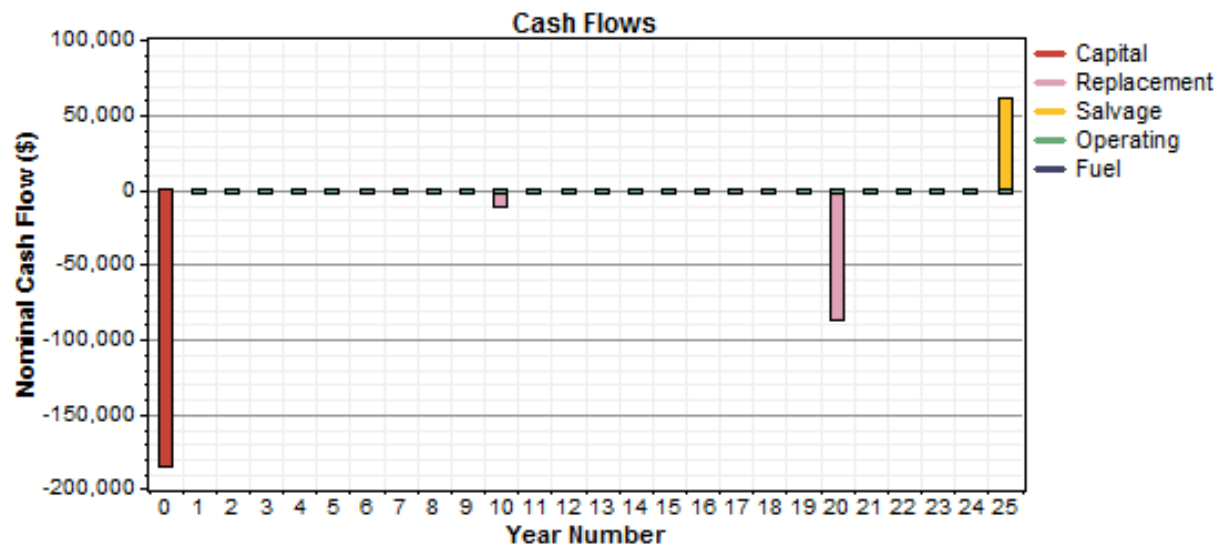
Cost summary

Total net present cost	\$ 226,661
Levelized cost of energy	\$ 0.341/kWh
Operating cost	\$ 3,198/yr



Net Present Costs

Component	Capital	Replacement	O&M	Fuel	Salvage	Total
	(\$)	(\$)	(\$)	(\$)	(\$)	(\$)
PV	85,074	0	10,968	0	0	96,042
FD6.4-5000-16	16,127	0	2,058	0	0	18,185
Hoppecke 24 OPzS 3000	75,792	23,632	9,689	0	-13,245	95,868
Converter	8,790	7,649	1,151	0	-1,024	16,566
System	185,783	31,281	23,866	0	-14,269	226,661

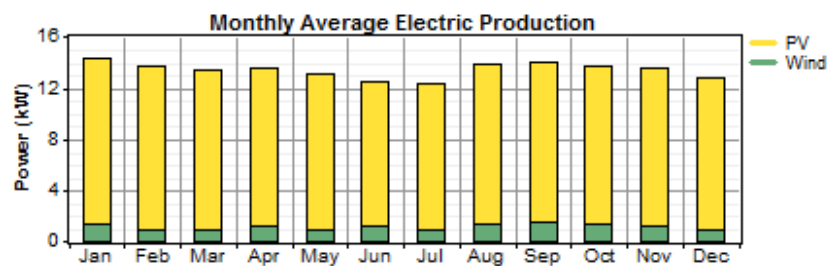


Annualized Costs

Component							Capital	Replacement	O&M	Fuel	Salvage	Total
							(\$/yr)	(\$/yr)	(\$/yr)	(\$/yr)	(\$/yr)	(\$/yr)
PV							6,655	0	858	0	0	7,513
FD6.4-5000-16							1,262	0	161	0	0	1,423
Hoppecke 24 OPzS 3000							5,929	1,849	758	0	-1,036	7,499
Converter	688	598	90	0	-80		1,296					
System	14,533	2,447	1,867	0	-1,116		17,731					

Electrical

Component	Production	Fraction
	(kWh/yr)	
PV array	107,193	91%
Wind turbine	10,231	9%
Total	117,424	100%



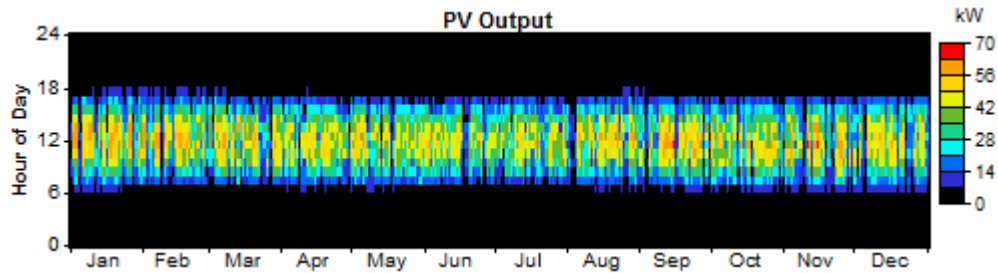
Load	Consumption	Fraction
	(kWh/yr)	
AC primary load	48,513	93%
DC primary load	3,416	7%
Total	51,929	100%

Quantity	Value	Units
Excess electricity	56,559	kWh/yr
Unmet load	32.4	kWh/yr
Capacity shortage	50.4	kWh/yr
Renewable fraction	1.000	

PV

Quantity	Value	Units
Rated capacity	66.0	kW
Mean output	12.2	kW
Mean output	294	kWh/d
Capacity factor	18.5	%
Total production	107,193	kWh/yr

Quantity	Value	Units
Minimum output	0.00	kW
Maximum output	63.9	kW
PV penetration	206	%
Hours of operation	4,378	hr/yr
Levelized cost	0.0701	\$/kWh



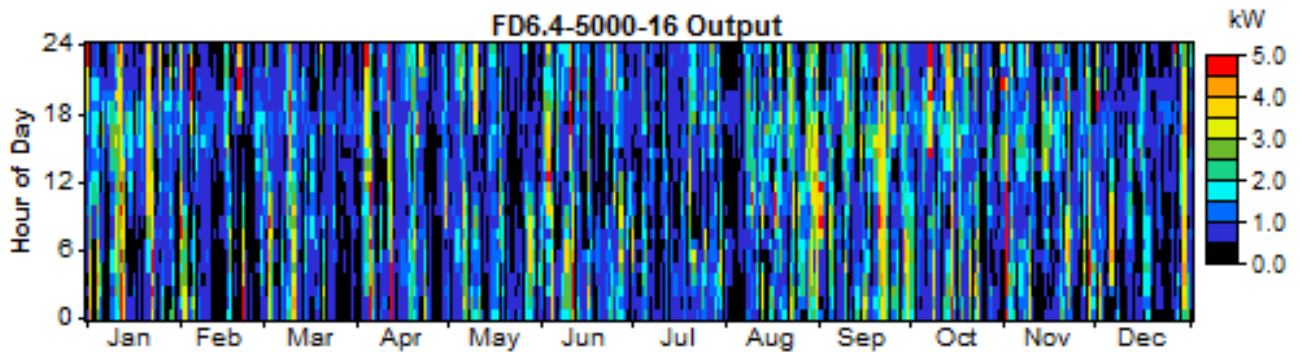
DC Wind Turbine: FD6.4-5000-16

Variable	Value	Units
Total rated capacity	5.00	kW
Mean output	1.17	kW
Capacity factor	23.4	%
Total production	10,231	kWh/yr

Variable	Value	Units
Minimum output	0.0357	kW
Maximum output	4.98	kW
Wind penetration	19.7	%
Hours of operation	8,760	hr/yr
Levelized cost	0.139	\$/kWh

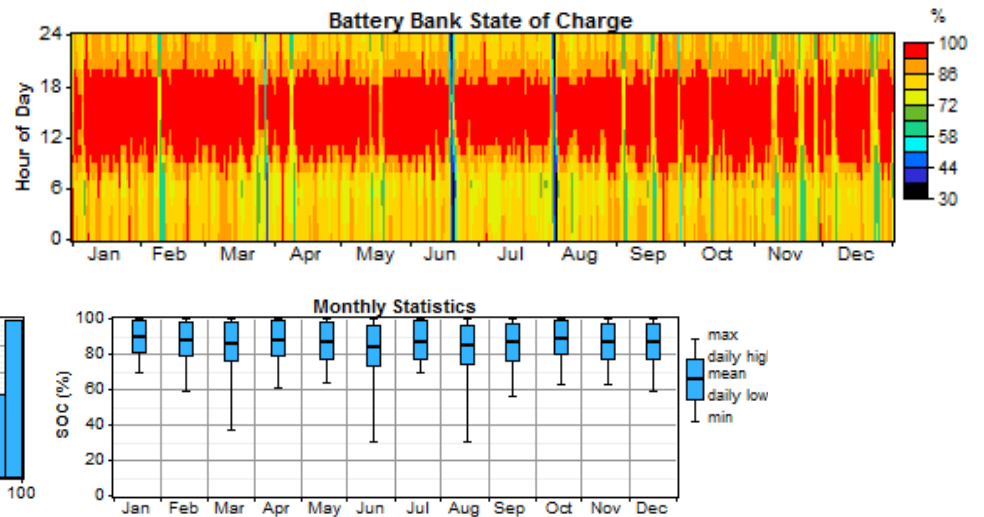
Quantity	Value	Units
Nominal capacity	288	kWh
Usable nominal capacity	202	kWh
Autonomy	34.0	hr
Lifetime throughput	489,408	kWh
Battery wear cost	0.167	\$/kWh
Average energy cost	0.000	\$/kWh

Quantity	Value	Units
Energy in	25,774	kWh/yr
Energy out	22,229	kWh/yr
Storage depletion	13.0	kWh/yr
Losses	3,533	kWh/yr
Annual throughput	23,970	kWh/yr
Expected life	20.0	yr

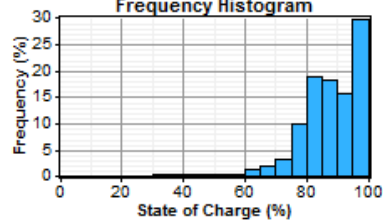


Battery

Quantity	Value
String size	24
Strings in parallel	2
Batteries	48
Bus voltage (V)	48



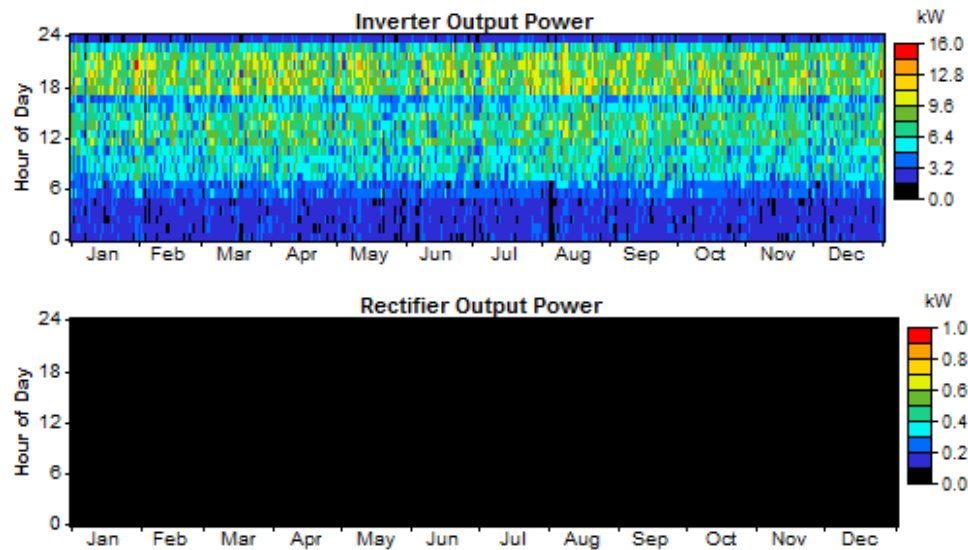
Frequency Histogram



Converter

Quantity	Inverter	Rectifier	Units
Capacity	15.0	15.0	kW
Mean output	5.5	0.0	kW
Minimum output	0.0	0.0	kW
Maximum output	15.0	0.0	kW
Capacity factor	36.9	0.0	%

Quantity	Inverter	Rectifier	Units
Hours of operation	8,757	0	hrs/yr
Energy in	53,903	0	kWh/yr
Energy out	48,513	0	kWh/yr
Losses	5,390	0	kWh/yr



Emissions

Pollutant	Emissions (kg/yr)
Carbon dioxide	0
Carbon monoxide	0
Unburned hydrocarbons	0
Particulate matter	0
Sulfur dioxide	0
Nitrogen oxides	0

SC2 – System Report

Sensitivity case

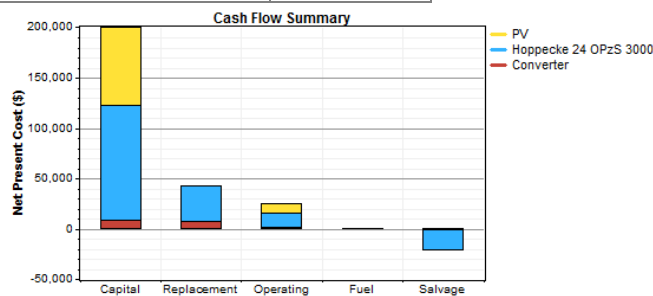
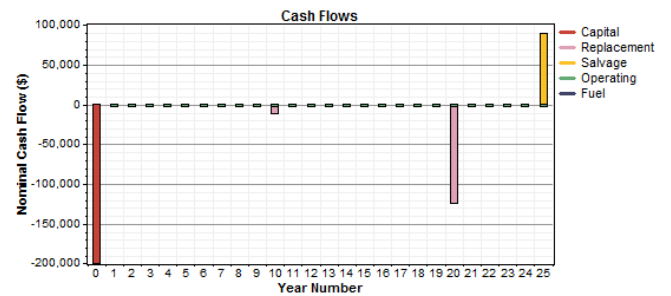
Solar Data Scaled Average: 5.13 kWh/m²/d
 Wind Data Scaled Average: 5.02 m/s
 WT FD8.0-10000 --10KW Hub Height: 12 m

System architecture

PV Array 60 kW
 Battery 72 Hoppecke 24 OPzS 3000
 Inverter 15 kW
 Rectifier 15 kW

Cost summary

Total net present cost	\$ 247,679
Levelized cost of energy	\$ 0.373/kWh
Operating cost	\$ 3,744/yr



Net Present Costs

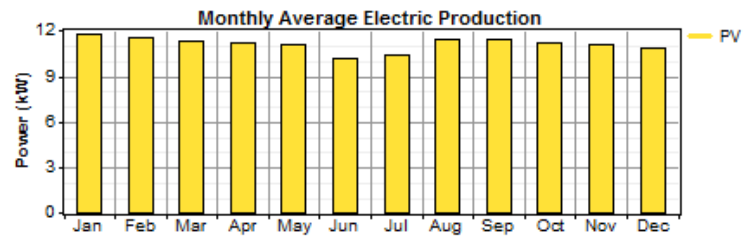
Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
PV	77,340	0	9,971	0	0	87,311
Hoppecke 24 OPzS 3000	113,688	35,448	14,533	0	-19,867	143,803
Converter	8,790	7,649	1,151	0	-1,024	16,566
System	199,818	43,098	25,655	0	-20,891	247,679

Annualized Costs

Component	Capital (\$/yr)	Replacement (\$/yr)	O&M (\$/yr)	Fuel (\$/yr)	Salvage (\$/yr)	Total (\$/yr)
PV	6,050	0	780	0	0	6,830
Hoppecke 24 OPzS 3000	8,893	2,773	1,137	0	-1,554	11,249
Converter	688	598	90	0	-80	1,296
System	15,631	3,371	2,007	0	-1,634	19,375

Electrical

Component	Production	Fraction
	(kWh/yr)	
PV array	97,448	100%
Total	97,448	100%



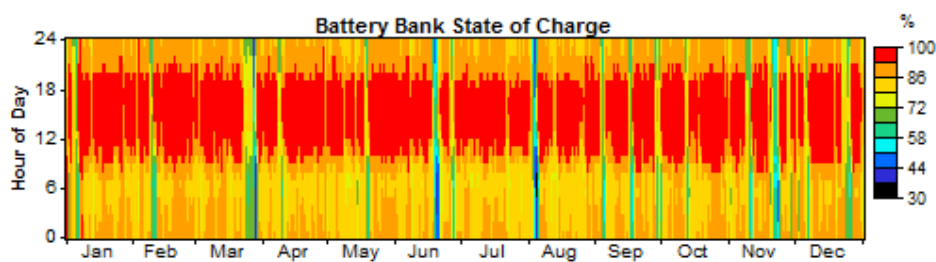
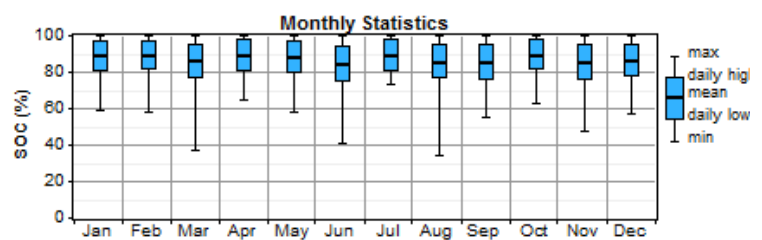
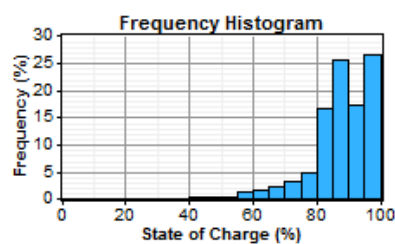
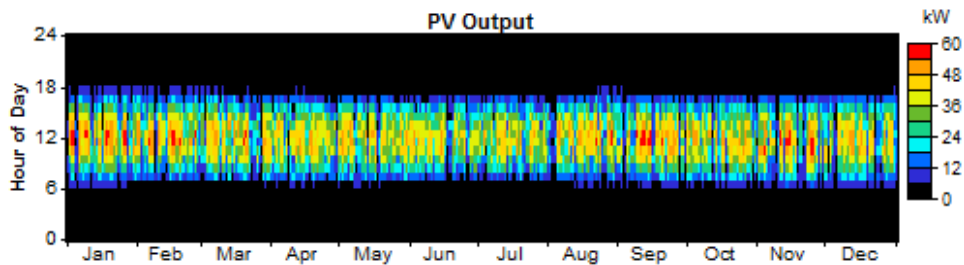
Load	Consumption	Fraction
	(kWh/yr)	
AC primary load	48,545	93%
DC primary load	3,416	7%
Total	51,961	100%

Quantity	Value	Units
Excess electricity	35,628	kWh/yr
Unmet load	0.291	kWh/yr
Capacity shortage	12.0	kWh/yr
Renewable fraction	1.000	

PV

Quantity	Value	Units
Rated capacity	60.0	kW
Mean output	11.1	kW
Mean output	267	kWh/d
Capacity factor	18.5	%
Total production	97,448	kWh/yr

Quantity	Value	Units
Minimum output	0.00	kW
Maximum output	58.1	kW
PV penetration	188	%
Hours of operation	4,378	hr/yr
Levelized cost	0.0701	\$/kWh



Battery

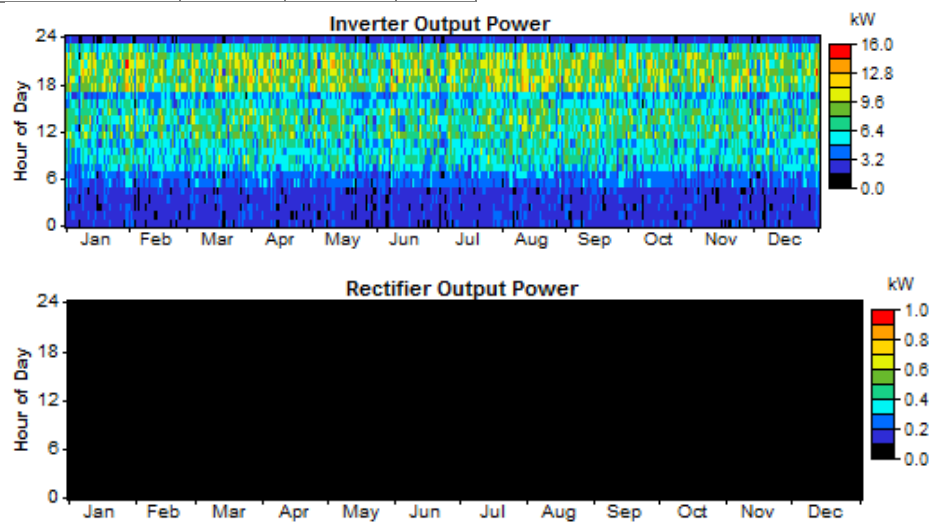
Quantity	Value
String size	24
Strings in parallel	3
Batteries	72
Bus voltage (V)	48

Quantity	Value	Units	Quantity	Value	Units
Nominal capacity	432	kWh	Energy in	32,728	kWh/yr
Usable nominal capacity	302	kWh	Energy out	28,262	kWh/yr
Autonomy	51.0	hr	Storage depletion	43.3	kWh/yr
Lifetime throughput	734,112	kWh	Losses	4,422	kWh/yr
Battery wear cost	0.167	\$/kWh	Annual throughput	30,476	kWh/yr
Average energy cost	0.000	\$/kWh	Expected life	20.0	yr

Converter

Quantity	Inverter	Rectifier	Units
Capacity	15.0	15.0	kW
Mean output	5.5	0.0	kW
Minimum output	0.0	0.0	kW
Maximum output	15.0	0.0	kW
Capacity factor	36.9	0.0	%

Quantity	Inverter	Rectifier	Units
Hours of operation	8,759	0	hrs/yr
Energy in	53,939	0	kWh/yr
Energy out	48,545	0	kWh/yr
Losses	5,394	0	kWh/yr



Emissions

Pollutant	Emissions (kg/yr)
Carbon dioxide	0
Carbon monoxide	0
Unburned hydrocarbons	0
Particulate matter	0
Sulfur dioxide	0
Nitrogen oxides	0

SC3 – System Report

Sensitivity case

Solar Data Scaled Average: 5.13 kWh/m²/d

Wind Data Scaled Average: 5.02 m/s

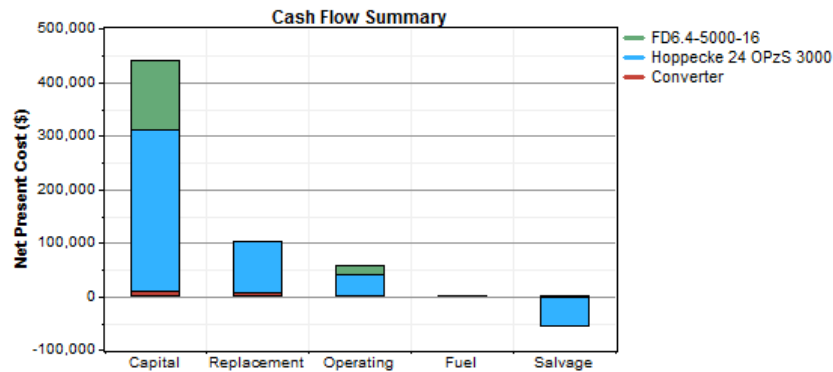
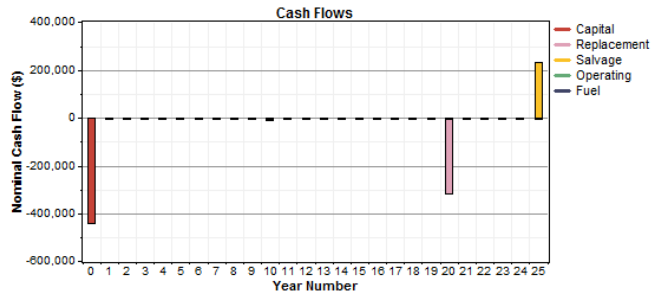
WT FD8.0-10000 --10KW Hub Height: 12 m

System architecture

Wind turbine	8 FD6.4-5000-16
Battery	192 Hoppecke 24 OPzS 3000
Inverter	15 kW
Rectifier	15 kW

Cost summary

Total net present cost	\$ 545,520
Levelized cost of energy	\$ 0.821/kWh
Operating cost	\$ 8,178/yr



Net Present Costs

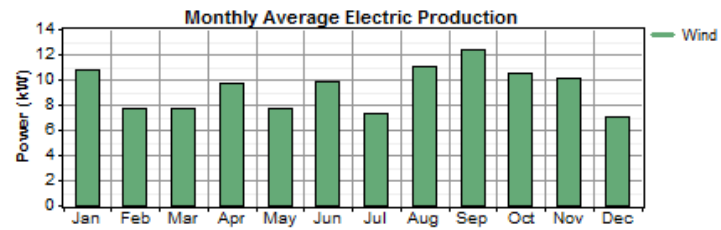
Component	Capital	Replacement	O&M	Fuel	Salvage	Total
	(\$)	(\$)	(\$)	(\$)	(\$)	(\$)
FD6.4-5000-16	129,016	0	16,465	0	0	145,481
Hoppecke 24 OPzS 3000	303,168	94,529	38,755	0	-52,978	383,474
Converter	8,790	7,649	1,151	0	-1,024	16,566
System	440,974	102,178	56,371	0	-54,002	545,520

Annualized Costs

Component	Capital	Replacement	O&M	Fuel	Salvage	Total
	(\$/yr)	(\$/yr)	(\$/yr)	(\$/yr)	(\$/yr)	(\$/yr)
FD6.4-5000-16	10,092	0	1,288	0	0	11,380
Hoppecke 24 OPzS 3000	23,716	7,395	3,032	0	-4,144	29,998
Converter	688	598	90	0	-80	1,296
System	34,496	7,993	4,410	0	-4,224	42,674

Electrical

Component	Production	Fraction
	(kWh/yr)	
Wind turbines	81,847	100%
Total	81,847	100%



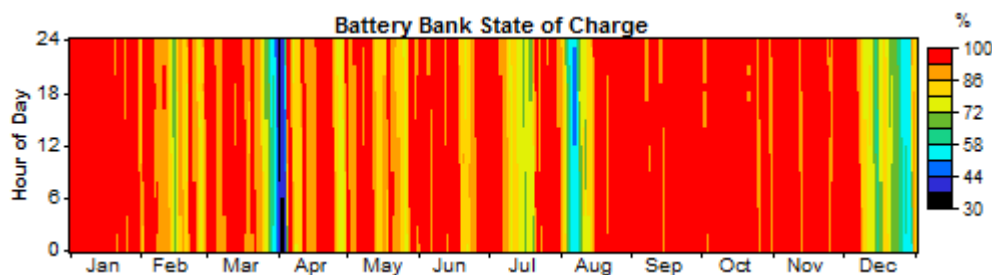
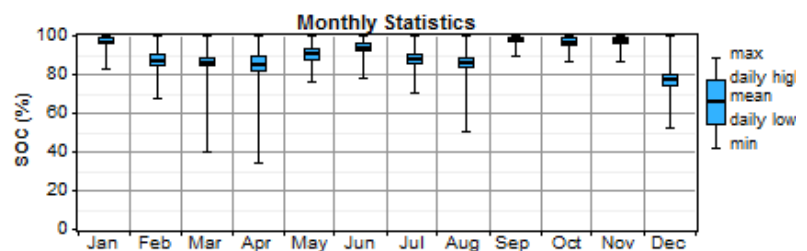
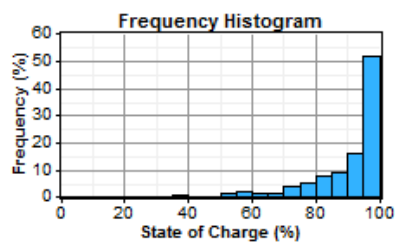
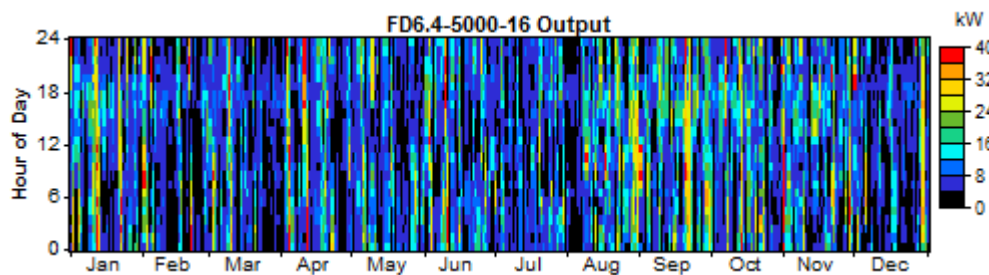
Load	Consumption	Fraction
	(kWh/yr)	
AC primary load	48,545	93%
DC primary load	3,416	7%
Total	51,961	100%

Quantity	Value	Units
Excess electricity	22,341	kWh/yr
Unmet load	0.292	kWh/yr
Capacity shortage	12.0	kWh/yr
Renewable fraction	1.000	

DC Wind Turbine: FD6.4-5000-16

Variable	Value	Units
Total rated capacity	40.0	kW
Mean output	9.34	kW
Capacity factor	23.4	%
Total production	81,847	kWh/yr

Variable	Value	Units
Minimum output	0.286	kW
Maximum output	39.8	kW
Wind penetration	158	%
Hours of operation	8,760	hr/yr
Levelized cost	0.139	\$/kWh



Battery

Quantity	Value
String size	24
Strings in parallel	8
Batteries	192
Bus voltage (V)	48

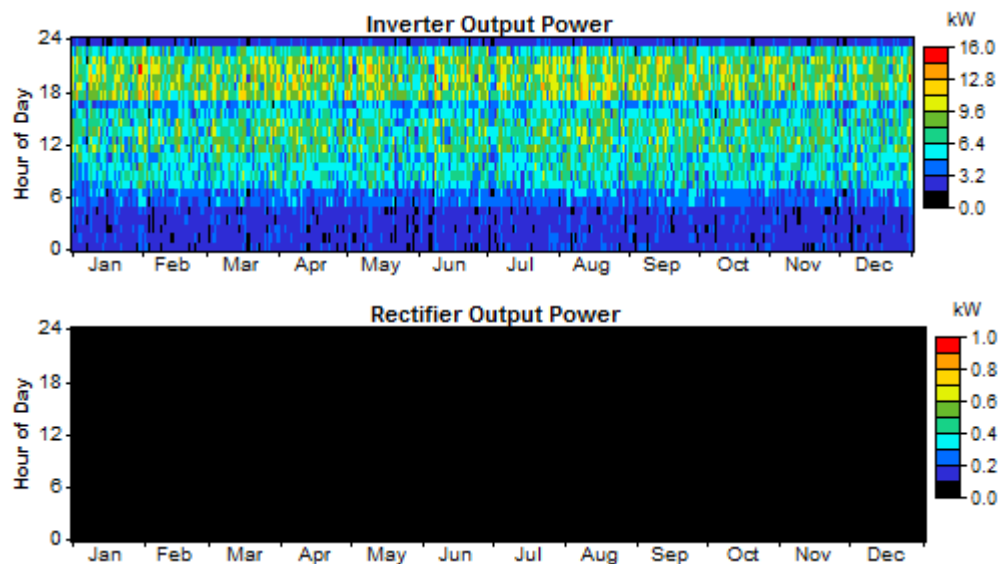
Quantity	Value	Units
Nominal capacity	1,152	kWh
Usable nominal capacity	806	kWh
Autonomy	136	hr
Lifetime throughput	1,957,632	kWh
Battery wear cost	0.167	\$/kWh
Average energy cost	0.000	\$/kWh

Quantity	Value	Units
Energy in	16,846	kWh/yr
Energy out	14,696	kWh/yr
Storage depletion	5.35	kWh/yr
Losses	2,145	kWh/yr
Annual throughput	15,847	kWh/yr
Expected life	20.0	yr

Converter

Quantity	Inverter	Rectifier	Units
Capacity	15.0	15.0	kW
Mean output	5.5	0.0	kW
Minimum output	0.0	0.0	kW
Maximum output	15.0	0.0	kW
Capacity factor	36.9	0.0	%

Quantity	Inverter	Rectifier	Units
Hours of operation	8,759	0	hrs/yr
Energy in	53,939	0	kWh/yr
Energy out	48,545	0	kWh/yr
Losses	5,394	0	kWh/yr



Emissions

Pollutant	Emissions (kg/yr)
Carbon dioxide	0
Carbon monoxide	0
Unburned hydrocarbons	0
Particulate matter	0
Sulfur dioxide	0
Nitrogen oxides	0